

Electrostatics and Current Electricity

CHAPTER 7

Heating Effects of Current

7.1–7.20

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7

Heating Effects of Current

HEATING EFFECTS OF CURRENT

When an electric current is passed through a conductor, it becomes hot and its temperature starts rising. This is known as **heating effect of current** or **Joule's heating effect**. Here electric energy converts into heat energy. Various appliances such as geyser, iron, heater, fuse wire, etc. work on this basis.

CAUSE OF HEATING

When current is passed through a conductor, the electrons start drifting toward the positive end. They gain additional kinetic energy (KE) apart from thermal KE. These electrons suffer collisions with atoms/ions more violently and transfer their KE to atoms/ions. It increases the amplitude of vibrations of ions/atoms. Thus, the average KE of vibrations of atoms/ions increases, which shows up in the form of increased temperature. Here, the electric energy supplied by the source of emf is converted into heat.

HEAT PRODUCED BY AN ELECTRIC CURRENT

Suppose a current I is flowing in a resistor of resistance R (figure). The amount of charge passed through the resistor in time t is $q = It$. Decrease in the potential of this charge is given by $V = IR$, and decrease in the potential energy of the charge is given by $qV = I^2Rt$.



This decrease in the energy will appear in the form of heat energy. So the electric energy produced in a resistor of resistance R in time t in which a current I is flowing is given by $H = I^2Rt$, which is Joule's law of heating.

Thus, Joule's law of heating states that the amount of heat produced in a conductor is directly proportional to (1) the square of the current, (2) the resistance of the conductor, and (3) time. Other forms of H are as follows:

$$H = I^2Rt = \frac{V^2}{R}t = VIt$$

Joule's heating effect is irreversible. The resistor will become hot (and not cool down), irrespective of the direction of current. As $H \propto I^2$, heating effect of current is common to both dc and ac. This is why instruments and appliances such as filament bulb, heater, geyser, press, toaster, etc. work on both dc and ac.

ELECTRIC POWER PRODUCED IN THE CIRCUIT

The electric power generated in the circuit is the energy produced in the resistor per unit time. Thus,

$$P = \frac{H}{t} = I^2R = \frac{V^2}{R} = VI$$

UNITS OF ELECTRIC ENERGY AND ELECTRIC POWER

Electric energy can be expressed in units such as J, cal, kWh, etc.

$$1 \text{ cal} = 4.18 \text{ J} \approx 4.2 \text{ J}$$

Relation between kWh and J:

$$1 \text{ kWh} = 1000 \text{ W} \times \text{h} = 1000 \text{ W} \times 3600 \text{ s} \\ = 3.6 \times 10^6 \text{ Ws} = 3.6 \times 10^6 \text{ J}$$

1 kWh is the energy consumed by an appliance of power 1 kW when it runs for 1 h.

Commercial unit: 1 kWh is one unit of electricity. To calculate the number of units, we can use the following relation:

$$\text{Number of units} = \frac{\text{watt} \times \text{hour}}{1000}$$

The energy dissipated in kWh can be calculated using the following relation:

$$E = \frac{V \text{ (in volt)} \times I \text{ (in ampere)} \times t \text{ (in hour)}}{1000}$$

Electric power can be expressed in units such as W, kW, MW, hp; they share the following relationships: $1 \text{ kW} = 10^3 \text{ W}$, $1 \text{ MW} = 10^6 \text{ W}$, $1 \text{ hp} = 746 \text{ W}$.

Important Points:

- If the resistances are connected in series, then using $P = I^2R$, the power developed will be higher in the resistor of higher value as current will be same in all resistors.
- If the resistances are connected in parallel, then using $P = V^2/R$, the power developed will be higher in the resistor of lower value as potential will be same across all resistors.

ILLUSTRATION 7.1

A series battery of 6 lead accumulators, each of emf 2.0 V and internal resistance 0.50Ω , is charged by a 100 V dc supply. What series resistance should be used in the charging circuit in order to limit the current to 8 A? Using the required resistor, obtain:

- the power supplied by the dc source
- the power dissipated as heat and
- chemical energy stored in the battery in 15 min.

Sol. We are given that emf of the charging supply, $\varepsilon_1 = 100 \text{ V}$
The total emf of the battery which is being charged,

$$\varepsilon_2 = 2.0 \times 6 = 12 \text{ V}$$

Total internal resistance of the battery, $r = 0.50 \times 6 = 3.0 \Omega$

when the battery is being charged, the emf of the battery (ε_2) acts in a direction opposite to that (ε_1) of the dc supply. Hence, effective emf in the charging circuit,

$$\varepsilon = \varepsilon_1 - \varepsilon_2 = 100 - 12 = 88 \text{ V}$$

Let R be the resistance required to limit the current (I) to 8.0 A. Clearly,

$$I = \frac{\varepsilon}{R + r} \quad \text{or} \quad 8 = \frac{88}{R + 3} \quad \text{or} \quad R = 8 \Omega$$

- (a) Power supplied by the dc source, $P = \varepsilon_1 I$

$$\text{Hence, } P = 100 \times 8 = 800 \text{ W}$$

- (b) Power dissipated as heat,

$$P' = I^2(R + r) = (8)^2(8 + 3) = 704 \text{ W}$$

- (c) Energy stored in the battery in 15 min (i.e., 15×60 s)

$$= (P - P')(15 \times 60) = (800 - 704) \times 900 = 86400 \text{ J}$$

ILLUSTRATION 7.2

A dry cell of emf 1.5 V and internal resistance 0.10Ω is connected across a resistor in series with a very low resistance ammeter. When the circuit is switched on, the ammeter reading settles to a steady value of 2.0 A.

- What is the steady rate of chemical energy consumption of the cell?
- What is the steady rate of energy dissipation inside the cell?
- What is the steady rate of energy dissipation inside the resistor?
- What is the steady power output of the source?

Sol.

- (a) Rate of chemical energy consumption of the cell is

$$EI = 1.5 \times 2 = 3 \text{ W}$$

- (b) Rate of energy dissipation inside the cell is

$$I^2 r = (2)^2 \times 0.1 = 0.4 \text{ W}$$

- (c) Rate of energy dissipation inside the resistor is

$$I^2 R = EI - I^2 r = 3 - 0.4 = 2.6 \text{ W}$$

- (d) Power output of the source $= I^2 R = 2.6 \text{ W}$

ILLUSTRATION 7.3

Two wires of same mass, having ratio of lengths 1:2, density 1:3, and resistivity 2:1, are connected one by one to the same voltage supply. The rate of heat dissipation in the first wire is found to be 10 W. Find the rate of heat dissipation in the second wire.

Sol. Given

$$\frac{l_1}{l_2} = \frac{1}{2}, \frac{d_1}{d_2} = \frac{1}{3}, \frac{\rho_1}{\rho_2} = \frac{2}{1}$$

$$m = A_1 l_1 d_1 = A_2 l_2 d_2$$

$$\therefore \frac{P_2}{P_1} = \frac{V^2 / R_2}{V^2 / R_1} = \frac{R_1}{R_2} = \frac{\rho_1 l_1 A_2}{\rho_2 l_2 A_1} = \frac{\rho_1 l_1^2 d_1}{\rho_2 l_2^2 d_2}$$

$$= \frac{2}{1} \times \left(\frac{1}{2}\right)^2 \times \frac{1}{3} = \frac{1}{6}$$

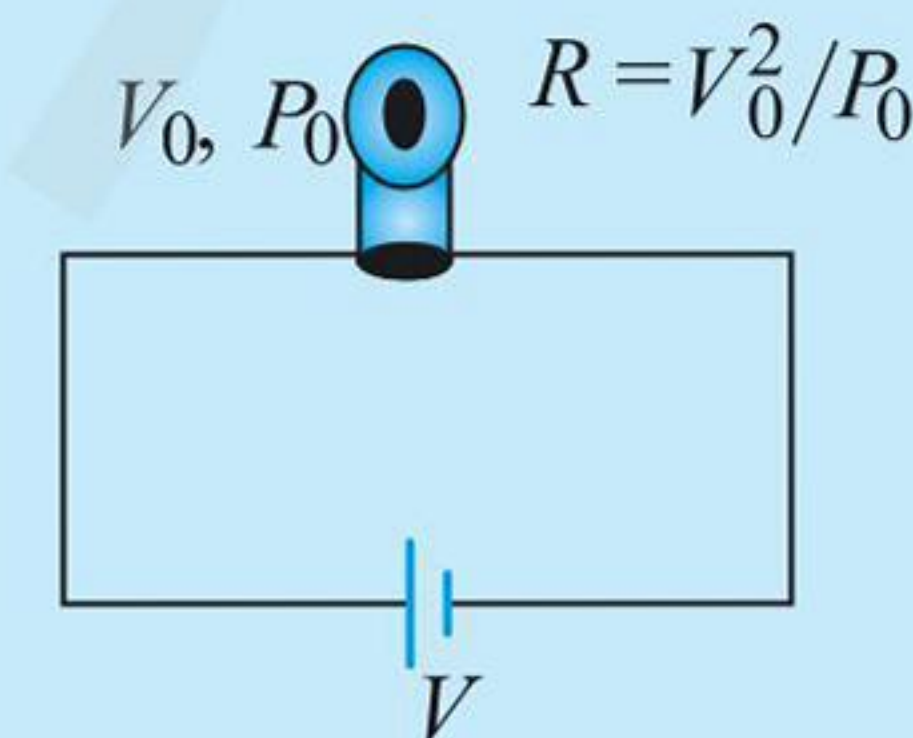
$$\text{or} \quad P_2 = \frac{P_1}{6} = \frac{10}{6} = \frac{5}{3} \text{ W}$$

Important Point:

Resistance of a bulb (or other appliances): Let a bulb be designed to operate on a voltage V_0 and its power indicated on it be P_0 (see figure). The resistance of the bulb is given by $R = V_0^2 / P_0$.

Now let a potential difference of V is applied across this bulb, then power consumed is given by

$$P = \frac{V^2}{R} = \left(\frac{V}{V_0}\right)^2 P_0$$



If $V = V_0$, then $P = P_0$. The above formula is very convenient to calculate the power consumption when the applied voltage is different from the specified one.

An electric appliance consumes the specified power P_0 only if it runs at the specified voltage V_0 . If the applied voltage V_A is greater than the specified voltage, the appliance may get damaged as in this situation, $I = V_A / R$ will exceed its current capacity $I_C = V_0 / R$. Further, if an appliance is made to run at a voltage lower than the specified, then true power consumption will be less than the specified value.

ILLUSTRATION 7.4

A 100 W bulb is designed to operate on a potential difference of 230 V.

- Find the resistance of the bulb.
- Find the current drawn by the bulb if it is operated at a potential difference for which it is designed.
- Find the current drawn and power consumed by the bulb if it is connected to a 200 V supply.

Sol. Power rating of the bulb is $P_0 = 100 \text{ W}$, voltage and rating of the bulb is $V_0 = 230 \text{ V}$.

- (a) Resistance of the bulb is

$$R = \frac{V_0^2}{P_0} = \frac{(230)^2}{100} = 529 \Omega$$

- (b) Current drawn is

$$I = \frac{V_0}{R} = \frac{230}{529} = \frac{10}{23} \text{ A}$$

- (c) $I = \frac{200}{529} \text{ A} \Rightarrow P = I^2 R = \left(\frac{200}{529}\right)^2 \times 529 = 75.6 \text{ W}$

ILLUSTRATION 7.5

A 500 W heating unit is designed to operate from a 200 V line. By what percentage will its heat output drop if the line voltage drops to 160 V? Find the heat produced by it in 10 min.

Sol. Actual power consumed is

$$P = \left(\frac{V}{V_0} \right)^2 P_0 = \left(\frac{160}{200} \right)^2 500 = 320 \text{ W}$$

Heat output drop is $500 - 320 = 180 \text{ W}$. Percentage heat drop is

$$\frac{180}{500} \times 100 = 36\%$$

Thus, heat produced in 10 min (600 s) is given by

$$H = 320 \times 600 = 192\,000 \text{ J} = 192 \text{ kJ}$$

Important Points:

- **Two bulbs connected in series:** Suppose two bulbs of same voltage rating V_0 and power ratings P_{01} and P_{02} are connected in series. Here $P_{01} > P_{02}$. Suppose potential V is applied across them as shown in figure.

Resistances of the bulbs are given by

$$R_1 = \frac{V_0^2}{P_{01}}, R_2 = \frac{V_0^2}{P_{02}}$$

Suppose powers produced in them are P_1 and P_2 , respectively. Then $P_1 = I^2 R_1$ and $P_2 = I^2 R_2$. Now

$$P_{01} > P_{02} \Rightarrow R_1 < R_2 \Rightarrow P_1 < P_2$$

It means the bulb having more power rating will consume less power when connected in series. So the total power produced is

$$\begin{aligned} P &= P_1 + P_2 = \frac{V^2}{R_1 + R_2} \\ &= \frac{V^2}{\frac{V_0^2}{P_{01}} + \frac{V_0^2}{P_{02}}} = \left(\frac{V}{V_0} \right)^2 \left(\frac{P_{01} P_{02}}{P_{01} + P_{02}} \right) \end{aligned}$$

If $V = V_0$, then

$$P = \frac{P_{01} P_{02}}{P_{01} + P_{02}} \quad \text{or} \quad \frac{1}{P} = \frac{1}{P_{01}} + \frac{1}{P_{02}}$$

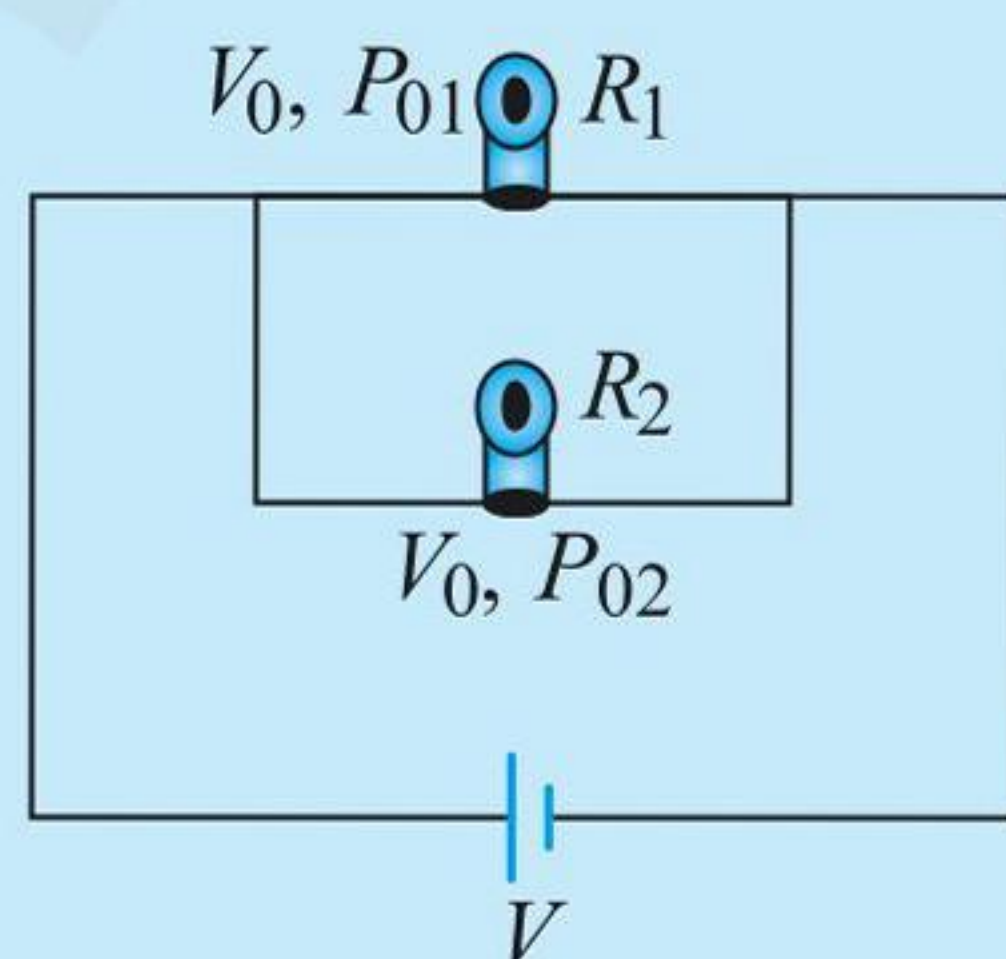
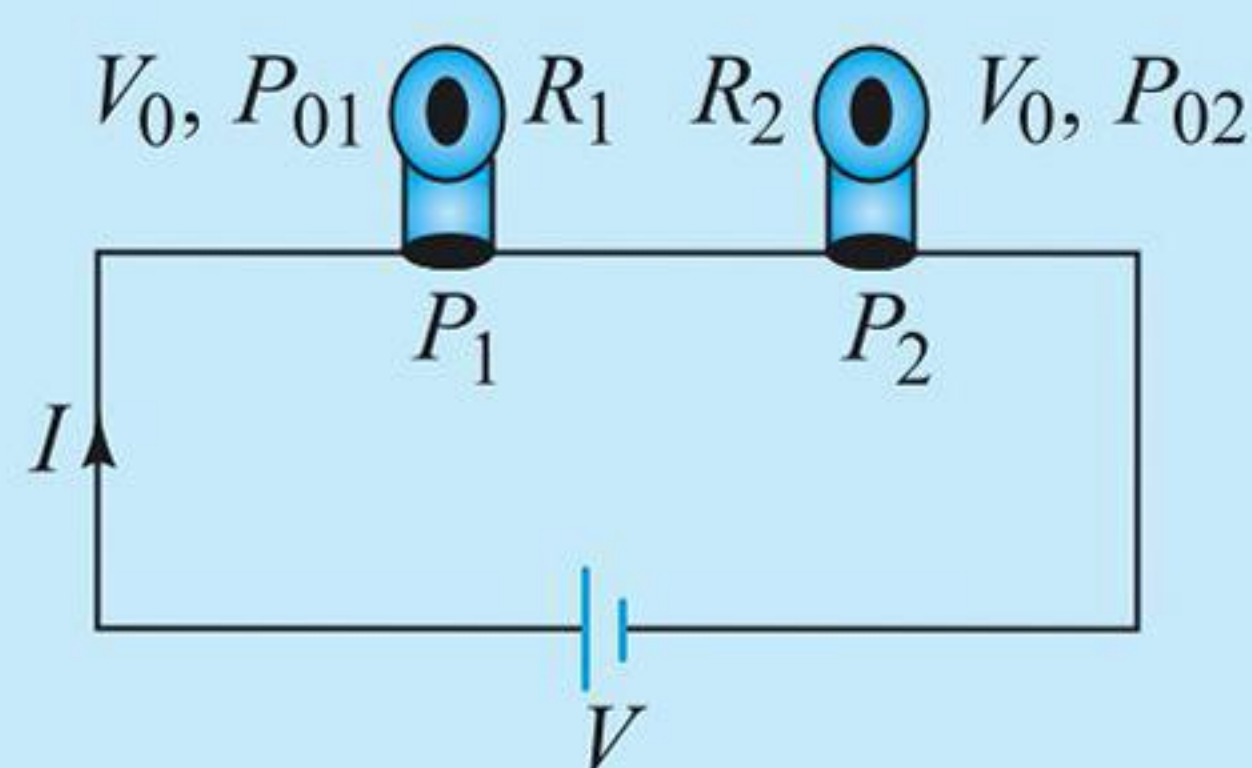
Note: If any one bulb in a series gets fused, then others will not glow.

- **Two bulbs connected in parallel:** Suppose two bulbs of same voltage rating V_0 and power ratings P_{01} and P_{02} are connected in parallel. Here $P_{01} > P_{02}$. Suppose potential V is applied across them as shown in figure. Resistances of the bulbs is

$$R_1 = \frac{V_0^2}{P_{01}}, R_2 = \frac{V_0^2}{P_{02}}$$

Let powers produced in them be P_1 and P_2 , respectively. Then $P_1 = V^2/R_1$ and $P_2 = V^2/R_2$.

$$P_{01} > P_{02} \Rightarrow R_1 < R_2 \Rightarrow P_1 > P_2$$



It means that the bulb having more power rating will consume more power when connected in parallel. So the total power produced is

$$\begin{aligned} P &= P_1 + P_2 = \frac{V^2}{R_1} + \frac{V^2}{R_2} = \frac{V^2}{V_0^2} P_{01} + \frac{V^2}{V_0^2} P_{02} \\ &= \left(\frac{V}{V_0} \right)^2 (P_{01} + P_{02}) \end{aligned}$$

If $V = V_0$, then $P = P_{01} + P_{02}$.

Note: If one of the bulbs in a parallel connection gets fused, then others will continue to glow.

ILLUSTRATION 7.6

Two bulbs are marked 220 V–100 W and 220 V–50 W.

- Which bulb will produce more illumination if they are connected in parallel to a 220 V supply?
- Which bulb will produce more illumination if they are connected in series to a 220 V supply?
- Also find the total power consumed by both the bulbs in each of the two parts above.

Sol. Given bulbs are 220 V–100 W and 220 V–50 W.

- In parallel combination of the bulbs, the more the power rating, the more the power produced.

The first bulb will produce more power.

- In series combination of the bulbs, the lesser the power rating, the more the power produced.

The second bulb will produce more power.

- In parallel combination of the bulbs,

Total power consumed,

$$P = P_{01} + P_{02} = 100 + 50 = 150 \text{ W.}$$

In series combination of the bulbs,

Total power consumed,

$$P = \frac{P_{01} P_{02}}{P_{01} + P_{02}} = \frac{100 \times 50}{150} = \frac{100}{3} \text{ W}$$

ILLUSTRATION 7.7

Two bulbs are rated 30 W–200 V and 60 W–200 V. They are connected with a 400 V power supply. Find which bulb will get fused if they are connected in (a) series and (b) parallel.

Sol. Resistance of the bulb, $R = \frac{V_0^2}{P_0}$

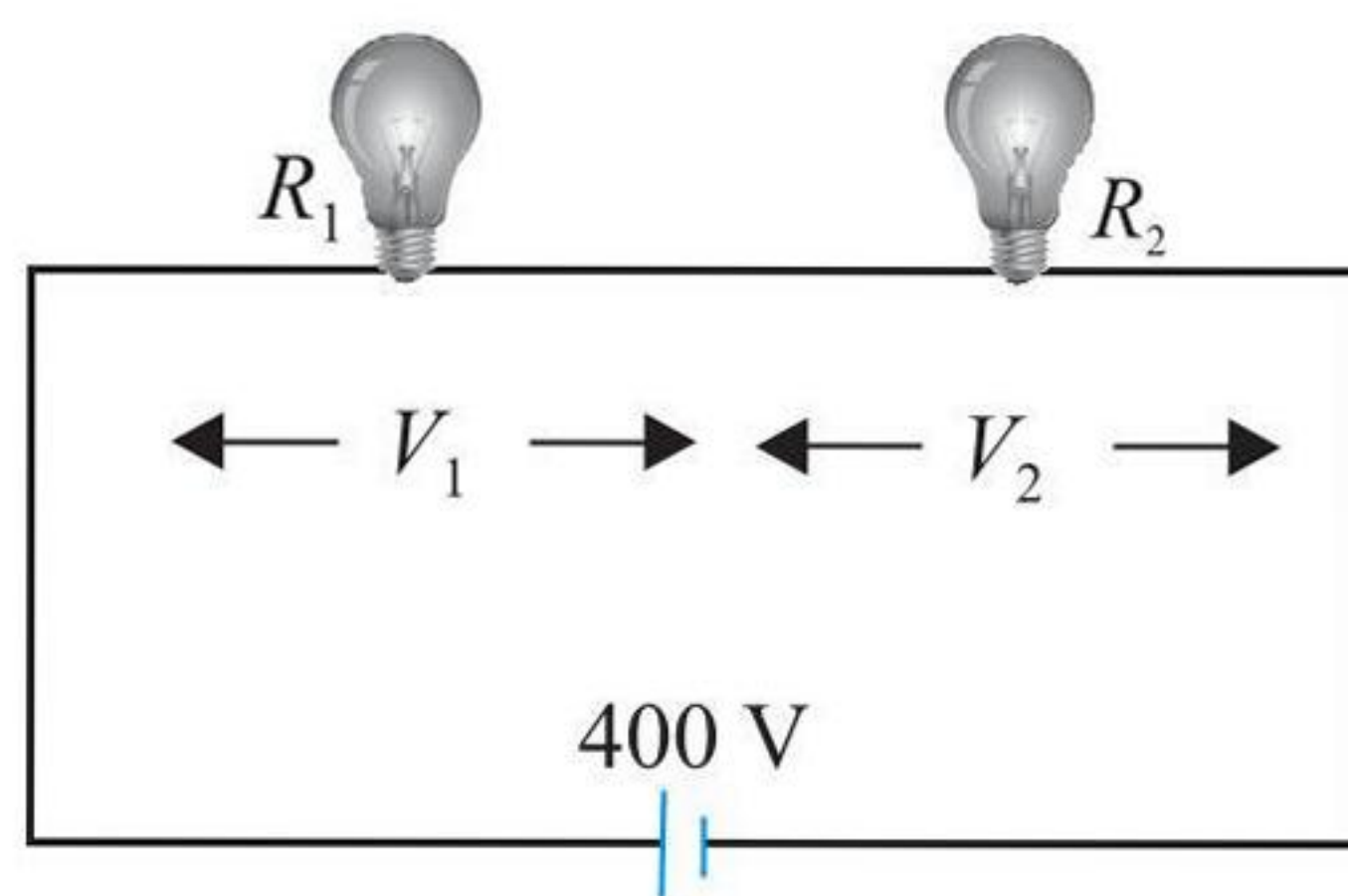
The resistance of the 1st bulb,

$$R_1 = \frac{(200)^2}{30} = \frac{4000}{3} \Omega$$

The resistance of the 2nd bulb,

$$R_2 = \frac{(200)^2}{60} = \frac{2000}{3} \Omega$$

(a) If bulbs are connected in series,



The potential difference across 1st bulb,

$$V_1 = V \left(\frac{R_1}{R_1 + R_2} \right)$$

$$V_1 = 400 \left(\frac{\frac{4000}{3}}{\frac{4000}{3} + \frac{2000}{3}} \right) = 400 \times \frac{2}{3} = \frac{800}{3} \text{ V} = 266.7 \text{ V}$$

The voltage across this bulb is greater than 200 V, so it will get fused.

The potential difference across 2nd bulb,

$$V_2 = 400 - 266.7 = 133.3 \text{ V}$$

The voltage across this bulb is less than 200 V, So it will not fuse.

(b) If bulbs are connected in parallel,

The potential across both the bulbs will be same and equal to 400 V.

The voltage across both the bulbs is greater than 200 V, So both will get fused.

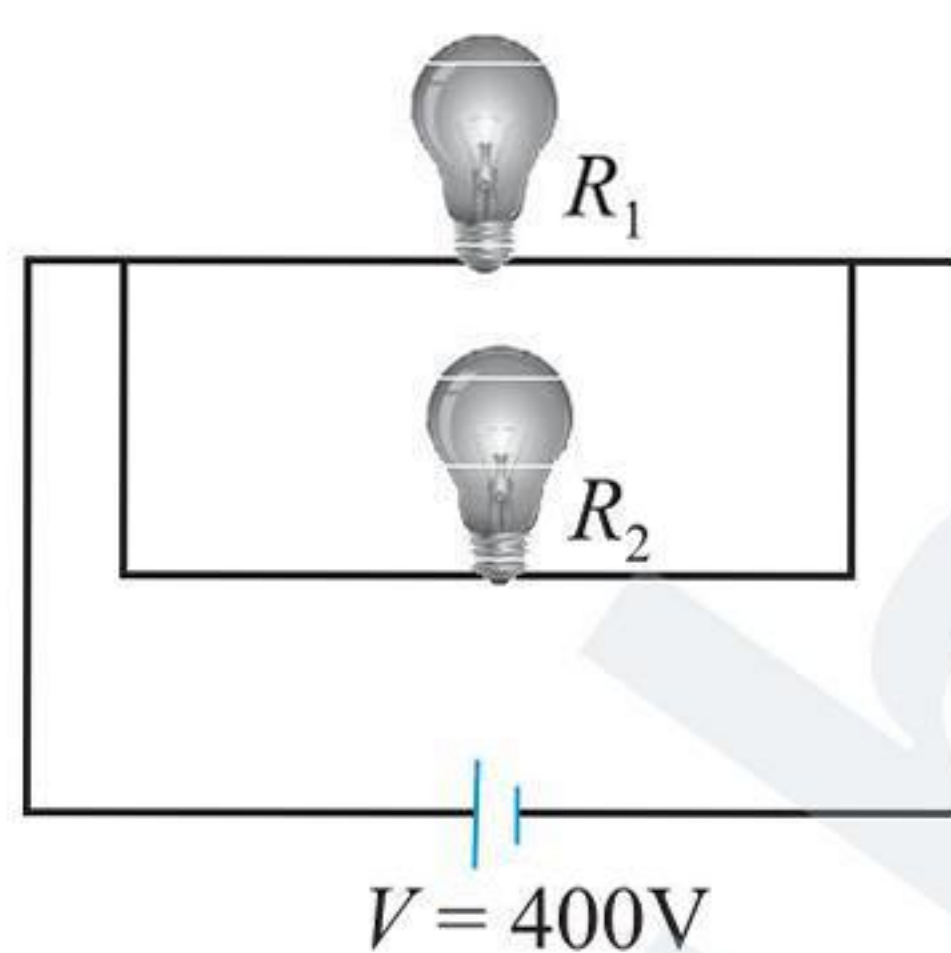


ILLUSTRATION 7.8

An electric tea kettle has two heating coils. When one of the coils is switched on, boiling begins in 6 min. When the other coil is switched on, boiling begins in 8 min. In what time will the boiling begin if both coils are switched on simultaneously (a) in series and (b) in parallel.

Sol. Let the power of first coil be P_1 and that of the second coil is P_2 . Let H be the amount of heat required to boil water. Then $H = P_1 t_1 = P_2 t_2$, where $t_1 = 6$ min and $t_2 = 8$ min.

(a) When the coils are connected in series,

$$P = \frac{P_1 P_2}{P_1 + P_2}$$

$$t = \frac{H}{P} = H \left[\frac{1}{P_1} + \frac{1}{P_2} \right] = H \left[\frac{t_1}{H} + \frac{t_2}{H} \right]$$

$$= t_1 + t_2 = 6 + 8 = 14 \text{ min}$$

(b) When the coils are connected in parallel,

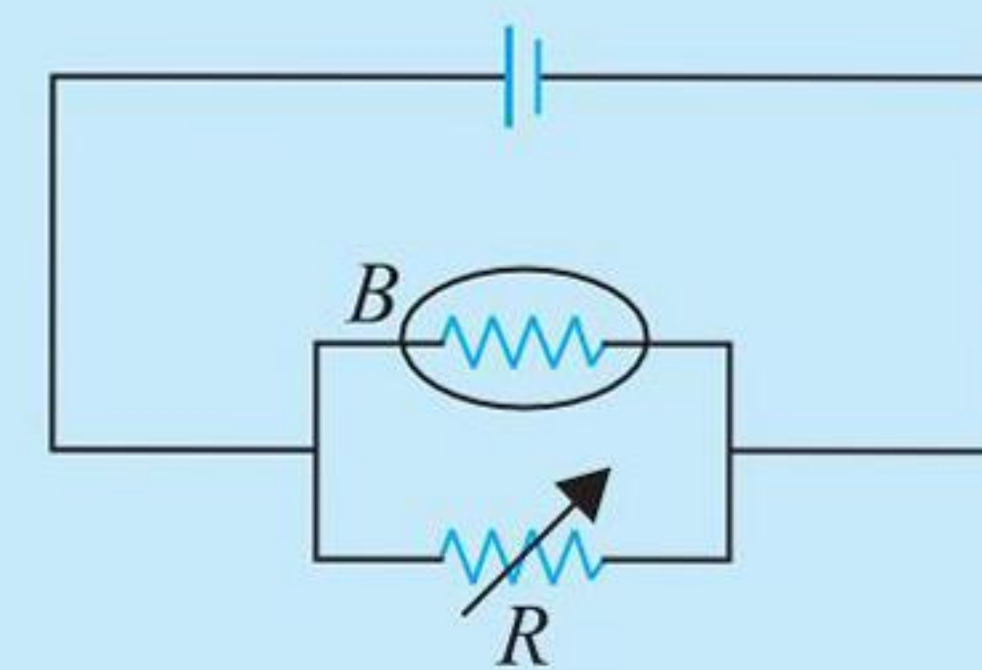
$$P = P_1 + P_2$$

$$t = \frac{H}{P} = \frac{H}{P_1 + P_2} = \frac{H}{\frac{H}{t_1} + \frac{H}{t_2}}$$

$$= \frac{t_1 t_2}{t_1 + t_2} = \frac{6 \times 8}{6 + 8} = 3.43 \text{ min}$$

ILLUSTRATION 7.9

A bulb B is connected to a source having constant emf and some internal resistance. A variable resistance R is connected in parallel to the bulb. If the resistance R is increased, how will it affect the



- Brightness of the bulb?
- Power spent by the source?

Sol. Let resistance of the bulb be R_b .

The equivalent of two parallel resistances is given by

$$\frac{1}{R_{eq}} = \frac{1}{R_b} + \frac{1}{R} \Rightarrow R = \frac{R R_b}{R + R_b} = \frac{R_b}{1 + \frac{R_b}{R}}$$

If R is increased, R_{eq} must also increase. This increases the overall resistance of the circuit and the current through the cell drops.

$$\text{As } i = \frac{\varepsilon}{r + R_{eq}} \text{ and } P_{\text{battery}} = \varepsilon i$$

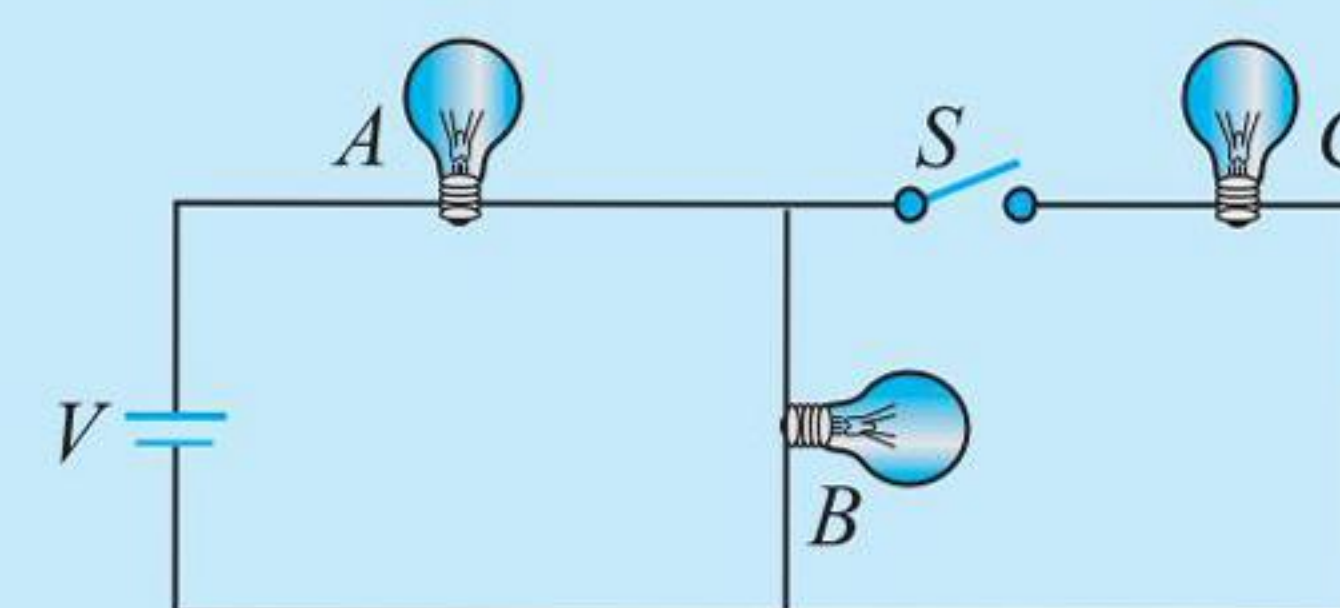
The potential drop across the bulb = potential difference across the battery

$$\text{Hence } V_{\text{battery}} = \varepsilon - ir = V_{\text{bulb}}$$

As the current i is decreasing, it means V_{battery} should increase. Hence the potential difference across the bulb will increase. It means power spent by the cell decrease. Potential drop across the internal resistance of the cell drops and the potential difference across the bulb increases. Hence, the bulb becomes brighter.

ILLUSTRATION 7.10

A and B are two identical bulbs of 40 W connected to a $V = 12$ volt cell. Switch S is closed to connect a third bulb C in the circuit. What happens to brightness of bulb A ? Answer for two cases.



- Bulb C is a very high wattage bulb.
 - Bulb C is a very low wattage bulb.
- All the three bulbs have rated voltage of 12 volt.

Sol. B and C are in parallel.

Therefore equivalent resistance of B and C will be smaller than that of B .

Hence, the overall resistance decreases.

Current through A will increase and brightness of A will increase. When C has high power rating, its resistance is very small. The equivalent of B and C is even smaller. Bulb A will nearly glow at its full brightness.

If C has low power rating, its resistance is too high.

Thus equivalent resistance of B and C will be only slightly less than that of B .

The brightness of A will increase slightly.

Important Points:

- Let a resistance R under a potential difference V dissipate power, then

$$P = \frac{V^2}{R}$$

So if the resistance is changed from R to R/n keeping V same, the power consumed will be

$$P' = \frac{V^2}{R/n} = n \frac{V^2}{R} = nP$$

That is, if for a given voltage, resistance is changed from R to R/n , power consumed changes from P to nP .

- If n equal resistances are connected in series with a voltage source, then power dissipated will be

$$P_s = \frac{V^2}{nR} \quad [\text{as } R_s = nR]$$

And if the same resistances are connected in parallel with the same voltage source,

$$P_p = \frac{V^2}{R/n} = \frac{nV^2}{R} \quad [\text{as } R_p = R/n]$$

$$\therefore \frac{P_p}{P_s} = n^2$$

$$\text{or } P_p = n^2 P_s$$

That is, power consumed by n equal resistors in parallel is n^2 times that of the power consumed in series, if V remains same.

CONCEPT APPLICATION EXERCISE 7.1

- A heater joined in series with a 50 W bulb is connected to the mains. If the 50 W bulb is replaced by a 100 W bulb, then will the heater now give more heat, less heat, or same heat? Why?
- An electric bulb rated 220 V and 60 W is connected in series with another electric bulb rated 220 V and 40 W. The combination is connected across a source of emf 220 V. Which bulb will glow more?
- We have a 30 W, 6 V bulb, which we want to glow by a supply of 120 V. What can be done for this?
- Two wires of the same material and having the same uniform area of cross section are connected in an electric circuit. The masses of the wires are m and $2m$, respectively. When a current I flows through both of them connected in series, then find the ratio of heat produced in them in a given time.
- Water boils in an electric kettle in 15 min after being switched on. Using the same main supply, should the length of the heating element be increased or decreased if the water is to be boiled in 10 min? Why?
- A house is fitted with certain numbers of 100 W, 230 V incandescent lamps. The power to the house is fed by a generator producing the power at 240 V. The resistance of the wires from the generator to the house is 2Ω . Find the maximum number of lamps that can be illuminated so that the voltage across none of the lamps drops below 230 V.

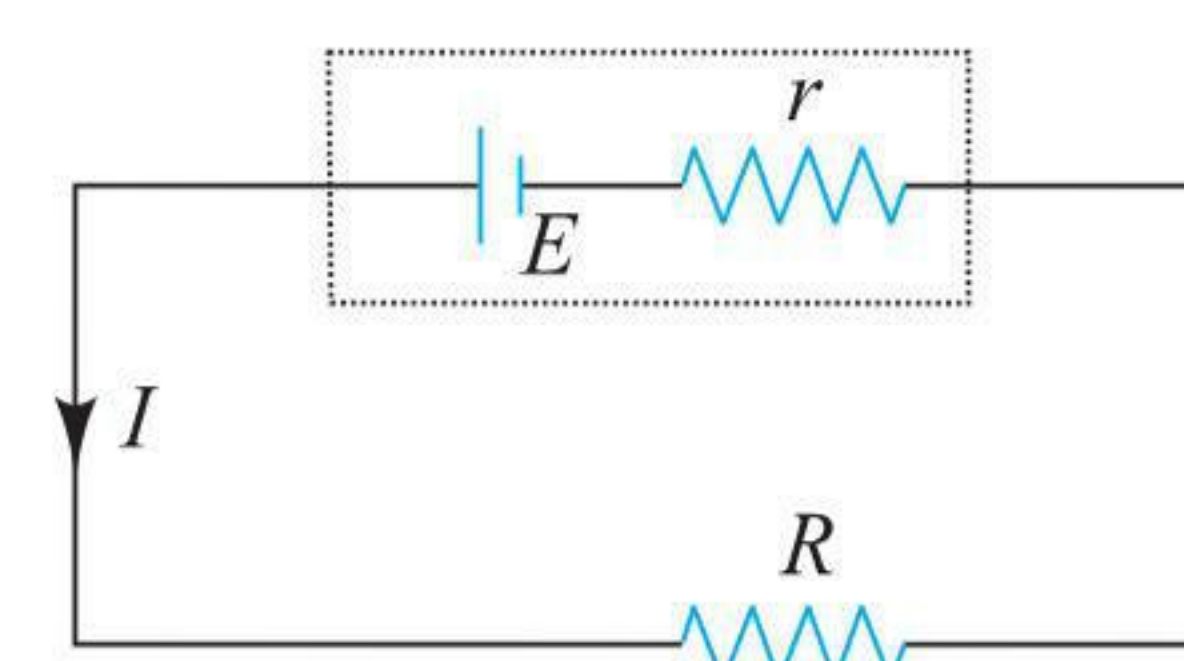
- A house is fitted with seven tubelights of rating 220 V, 40 W each, two bulbs of rating 220 V, 60 W each, five fans each drawing a current of 0.4 A at 220 V, and a heater of resistance 48.4Ω . The main line power supplied to the house is at 220 V. Calculate the bill for the month of January if tubelights and bulbs are used for 6 h daily, fans for 1 h daily, and heater for 10 h daily. The electricity is to cost ₹2 per unit.
- Two bulbs are marked 200 V, 300 W and 200 V, 600 W, respectively. The bulbs are connected in series and the combination is connected to a 200 V supply.
 - Which bulb will produce more illumination?
 - Find the total power consumed by both the bulbs.
 - Find the total power consumed if both the bulbs were connected in parallel.
- A 25 W and a 100 W bulb are joined in series and connected to the mains. Which bulb will glow brighter?

ANSWERS

- Heater will give more heat
- The bulb rates 220 V and 40 W will glow more
- 22.8Ω resistance should be connected in series with bulb
- 1 : 2 5. Decreases 6. 11 7. ₹ 796.08
- (a) 1st bulb (b) 200 W (c) 900 W 9. 25 W bulb

MAXIMUM POWER TRANSFER THEOREM

Suppose we want to find for what value of external resistance the maximum power will be drawn from a battery. For this, in the shown network (figure), let the power developed in resistance R be



$$P = I^2 R = \frac{E^2}{(R + r)^2} R \quad \left(\text{as } I = \frac{E}{R + r} \right)$$

Now, for $dP/dR = 0$ (since P will be maximum if $dP/dR = 0$)

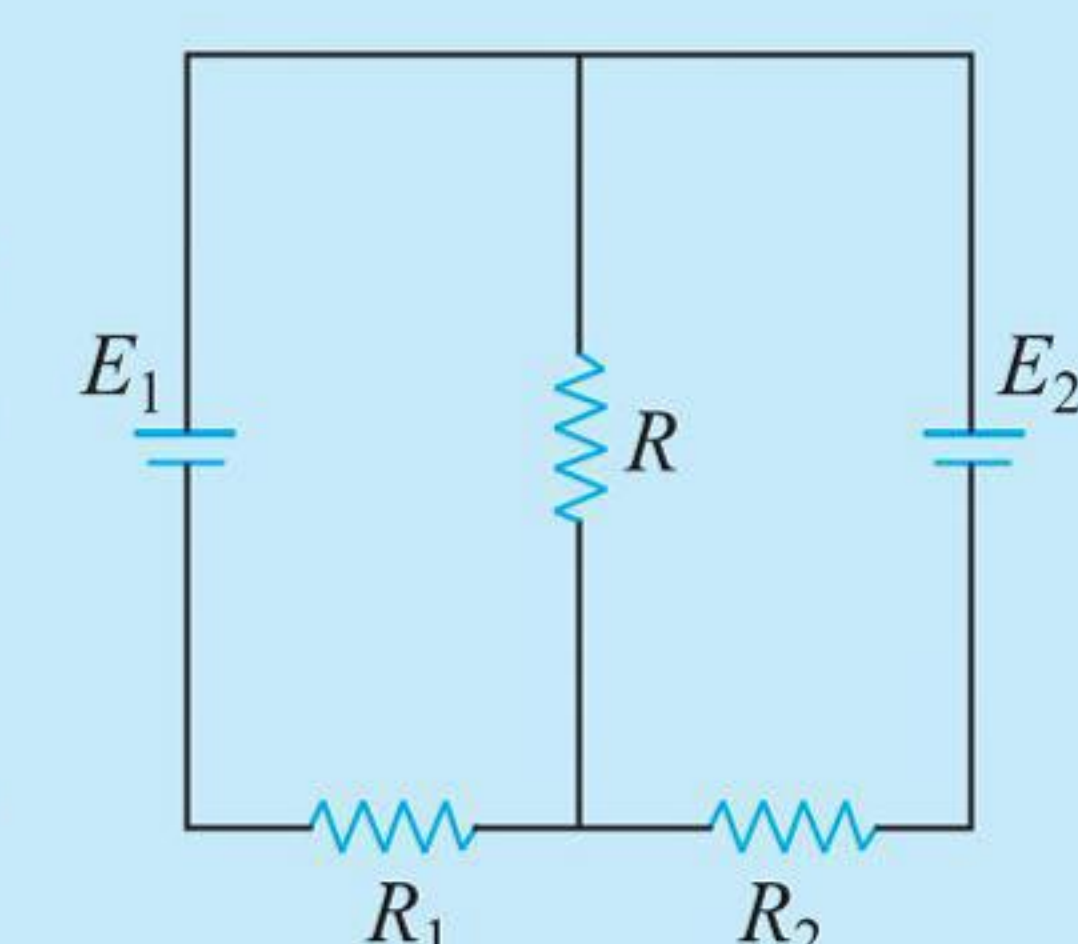
$$E^2 \frac{(R + r)^2 - 2(R)(R + r)}{(R + r)^4} = 0$$

$$\text{or } (r + R) = 2R \quad \text{or } r = R$$

It means the power output is maximum, when the external resistance equals the internal resistance i.e., when $R = r$.

ILLUSTRATION 7.11

In the circuit shown in figure, the emfs of batteries are E_1 and E_2 which have internal resistances R_1 and R_2 . At what value of the resistance R will the thermal power generated in it be the highest? What is it?



Sol. Here two batteries are in parallel. The thermal power generated in R will be maximum when, total internal resistance of the battery = total external load resistance. The total internal resistance of equivalent battery,

$$r = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2}$$

Hence for maximum power, $r = R = \frac{R_1 R_2}{R_1 + R_2}$

$$E_{\text{eq}} = \frac{\left(\frac{E_1}{R_1} + \frac{E_2}{R_2}\right)}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} = \left(\frac{E_1 R_2 + E_2 R_1}{R_1 + R_2}\right)$$

Net resistance in the circuit, $R_{\text{net}} = \frac{2R_1 R_2}{R_1 + R_2}$

$\therefore$ The current in the circuit, $i = \frac{E_{\text{eq}}}{R_{\text{net}}} = \frac{E_1 R_2 + E_2 R_1}{2R_1 R_2}$

Maximum power through R , $P_{\text{max}} = i^2 R = \frac{(E_1 R_2 + E_2 R_1)^2}{4R_1 R_2 (R_1 + R_2)}$

SOME APPLICATIONS

Fusing of bulb when it is switched on: Usually filament bulbs get fused when they are switched on. This is because with the rise in temperature, the resistance of the bulb increases and becomes constant in steady state. So the power consumed by the bulb (V^2/R) initially is more than that in steady state and hence the bulb glows more brightly in the beginning and may get fused.

ILLUSTRATION 7.12

Two wires made of tinned copper having identical cross section ($=10^{-6} \text{ m}^2$) and lengths 10 and 15 cm are to be used as fuses. Show that the fuses will melt at the same value of current in each case.

Sol. The temperature of the wire rises to a certain steady temperature when the heat produced per second by the current just becomes equal to the rate of loss of heat from its surface. Heat produced per second by the current is

$$P = I^2 R = I^2 \left(\frac{\rho l}{\pi r^2} \right) = \frac{I^2 \rho l}{\pi r^2} \quad \dots(i)$$

where l is the length, r is the radius, and ρ is the specific resistance. Let H be the heat lost per second per unit surface area of the wire. If we neglect the loss of heat from the end faces of the wire, then heat lost per second by the wire is $H \times$ surface area of wire $P' = H \times 2\pi r l$ $\dots(ii)$

At steady state temperature, from (i) and (ii)

$$H \times 2\pi r l = \frac{I^2 \rho l}{\pi r^2}$$

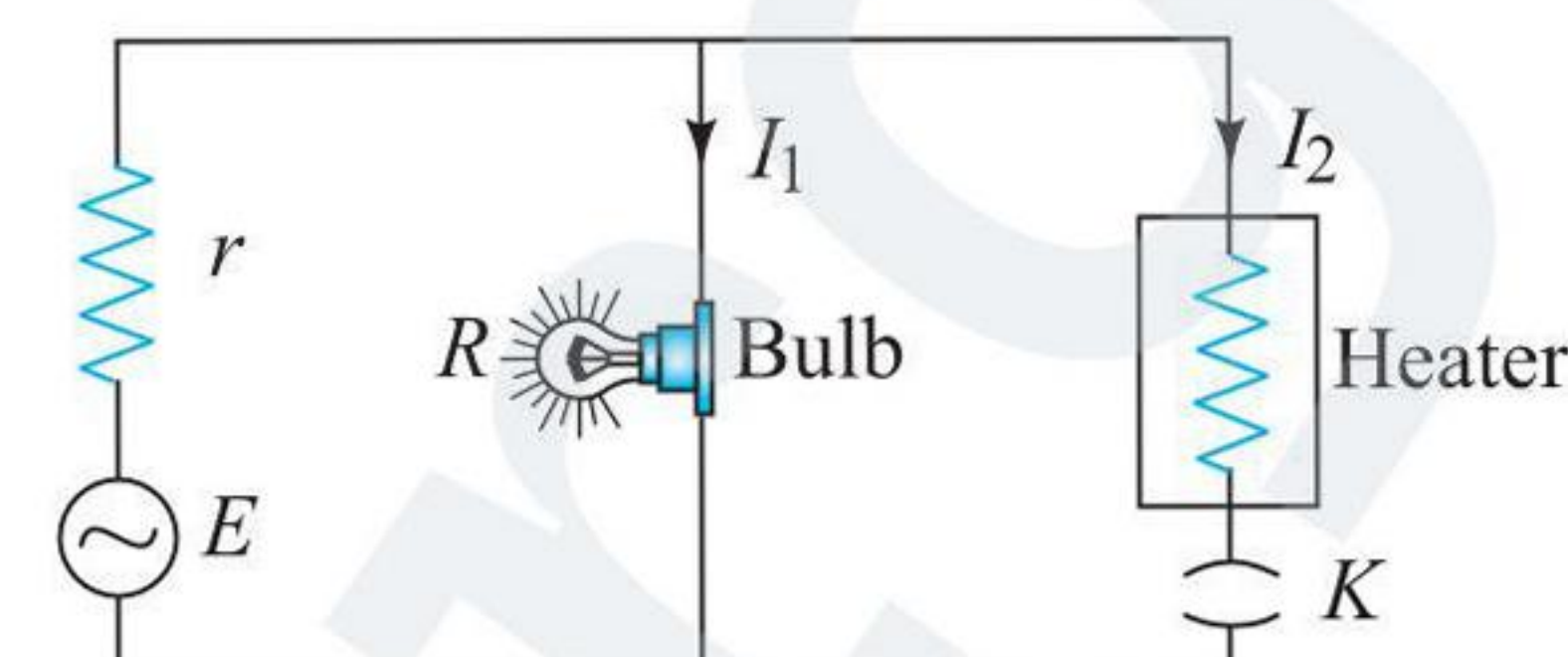
$$\text{or } H = \frac{I^2 \rho}{2\pi^2 r^3} \quad \dots(iii)$$

From Eq. (iii) we note that the rate of loss of heat (H), which, in turn, depends on the temperature of the wire, is independent of the length of the wire. Hence, the fuses of two wires of the same values of r and ρ but of different lengths will melt for the same value of current in each case.

Decrease in the brightness of bulb after long use: Also due to evaporation of metal from the filament (which deposits as black

substance on the inner side of glass wall), the filament of the bulb becomes thinner and thinner with use. This increases the resistance ($R = \rho L/\pi r^2$) of the bulb, and as $P = V^2/R$, the brightness of the light emitted by a bulb decreases gradually with time.

Decrease in brightness of a bulb in a room when a heavy current appliance is switched on: As shown in figure, if the bulb draws a current I_1 from the source, then the terminal voltage of source is $V = (E - I_1 r)$.



So the power consumed by the bulb is

$$P = \frac{V^2}{R} = \frac{(E - I_1 r)^2}{R}$$

When a heavy current appliance such as motor, heater, or geyser is switched on, it will draw a heavy current, say I_2 , from the source so that the terminal voltage becomes

$$V' = [E - (I_1 + I_2)r] = (V - I_2 r) \quad (< V)$$

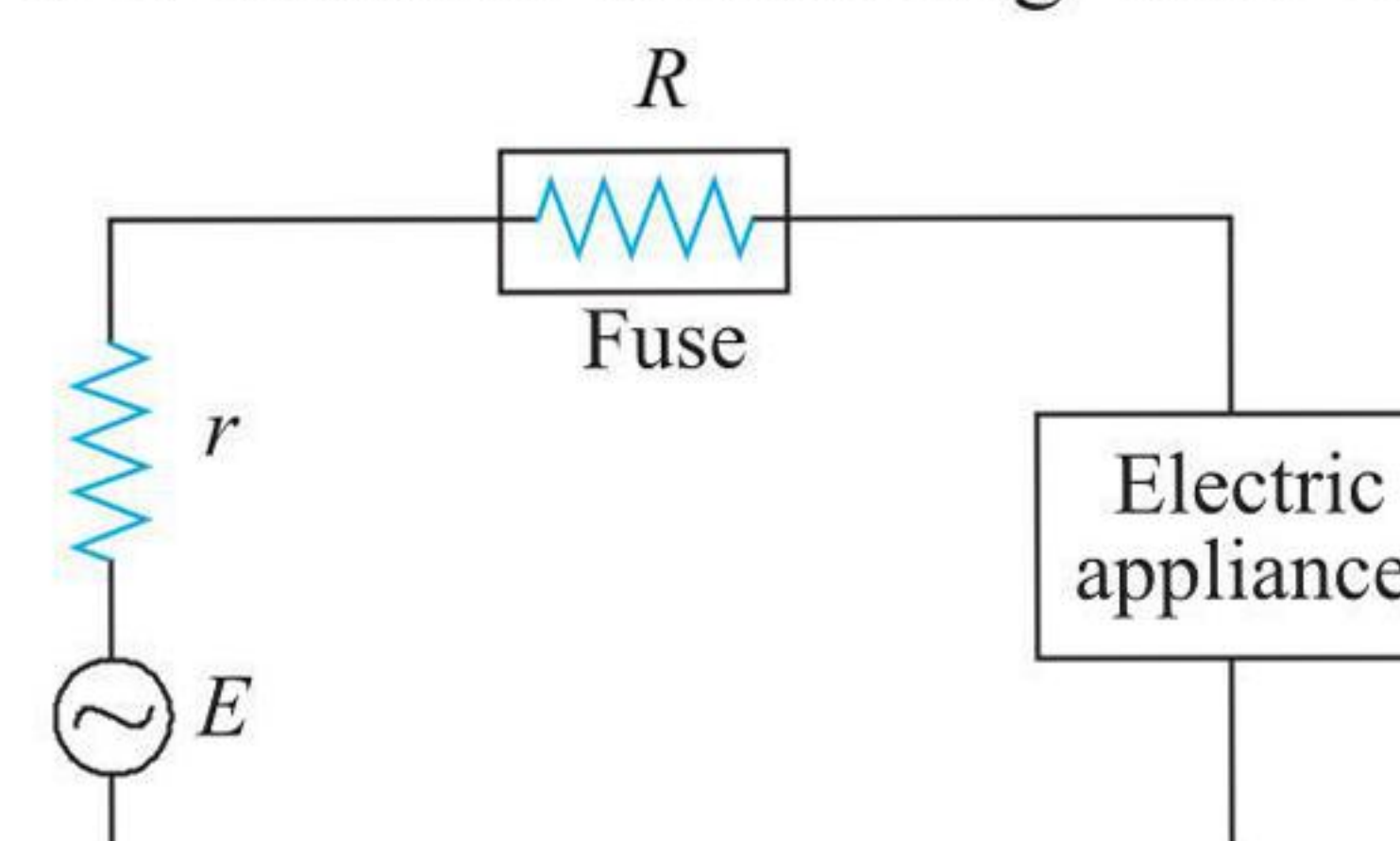
Hence the power consumed by the bulb will now be

$$P' = \frac{V'^2}{R} = \frac{(V - I_2 r)^2}{R} < P$$

So the brightness of the bulb decreases.

Note: If the source is ideal, i.e., $r = 0$, $V' = V = E$, then $P' = P$, i.e., there will be no change in the brightness of the bulb, if the source is ideal.

Fuse and its action: A fuse is a metallic conducting wire of 75% Pb and 25% Sn with low melting point and higher resistance and is in series with an appliance (figure). It is a safety device that protects the appliance from getting damaged, by melting and opening the circuit, if the current in the circuit exceeds a specific predetermined value, called current capacity.



Long distance power transmission: When power is transmitted through a power line of resistance R , power loss will be $I^2 R$. Now, let power P_0 is transmitted at voltage V , then $P_0 = VI$, i.e.,

$$I = \frac{P_0}{V}$$

So power loss is

$$\frac{P_0^2}{V^2} \times R$$

Since for a given power and line, P_0 and R are constant, power loss is proportional to $1/V^2$.

So if power is transmitted at high voltage, power loss will be small and vice versa, for example, power loss at 22 kV is 10^{-4} times than that at 220 V. This is why long distance power transmission is carried out at high voltage.

ILLUSTRATION 7.13

A line having a total resistance of $0.2\ \Omega$ delivers 10 kW at 220 V to a small factory. Calculate the efficiency of transmission.

Sol. The current through transmission line,

$$I = \frac{10,000}{220} = \frac{500}{11}\text{ A}$$

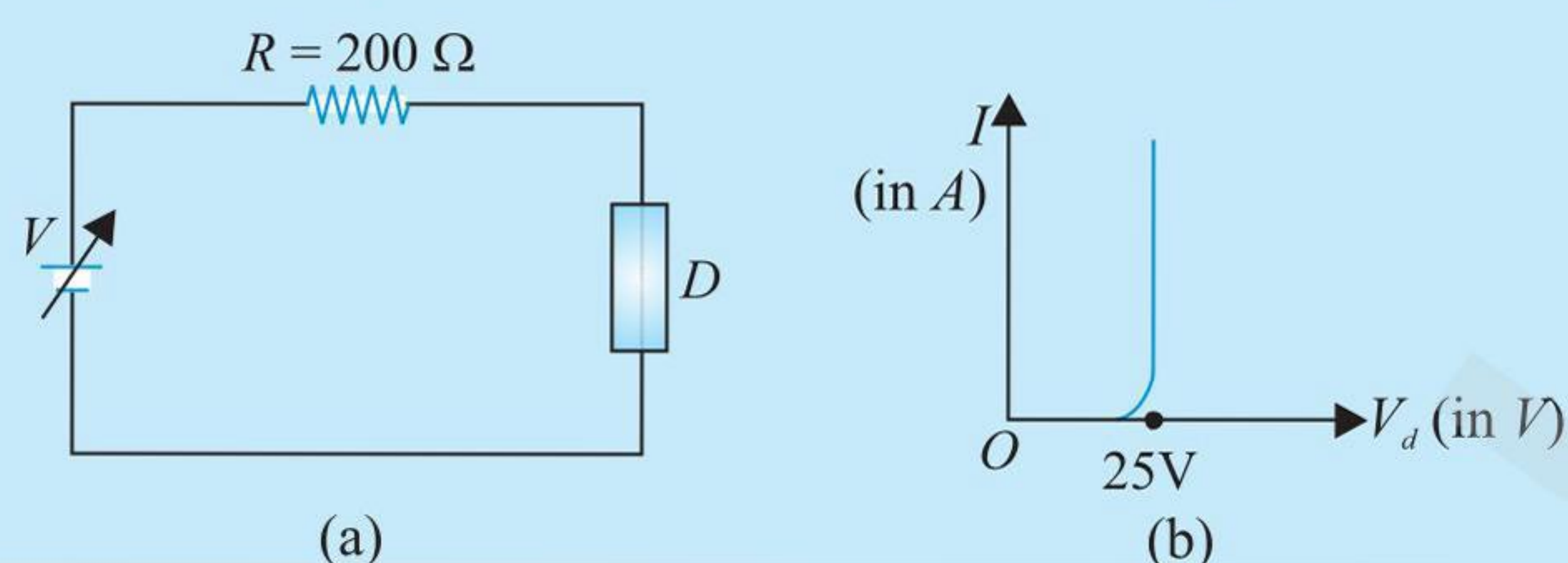
$$\text{Loss} = I^2 r = \left(\frac{500}{11}\right)^2 \times 0.2 = 413.22$$

So efficiency is

$$\eta = \frac{10,000}{10,000 + 413.22} \times 100 = 96\%$$

ILLUSTRATION 7.11

An electric device is connected with a source of variable emf (V) and a resistance ($R = 200\ \Omega$) in series as shown in Fig. (a). The current–voltage characteristic of an electric device is as shown in Fig. (b). The device gets damaged if power dissipated in it exceeds 5 watt . What is possible range of V for which the device remains operational (i.e., consumes some power) and safe.



Sol. From current–voltage characteristic graph it can be observed that the device conducts only when voltage across it is 25 V . Hence V should be greater than 25 V .

Maximum current through the device

$$I_{\max} = \frac{P_{\max}}{25\text{ volt}} = \frac{5\text{ Watt}}{25\text{ V}} = \frac{1}{5}\text{ A}$$

From the given circuit, $V = 200I + 25$

$$\text{Current in the circuit, } I = \frac{V - 25}{200}$$

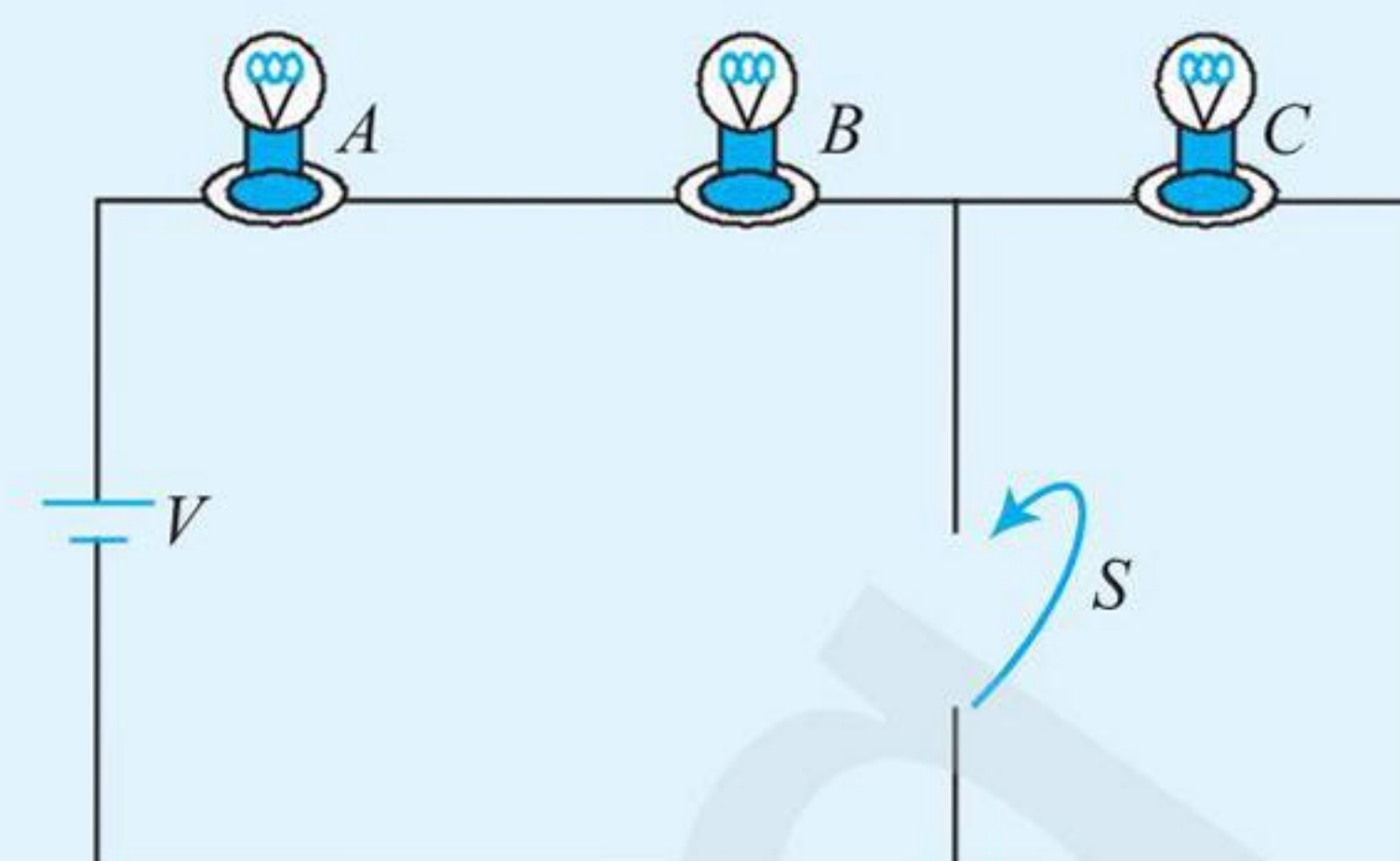
$$\text{But } I \leq \frac{1}{5}\text{ A}$$

$$\therefore \frac{V - 25}{200} \leq \frac{1}{5}$$

$$V - 25 \leq 40 \Rightarrow V = 45\text{ volt}$$

Hence required range should be

2. A series circuit consists of three identical lamps connected to a battery as shown in figure.



When the switch S is closed, what happens

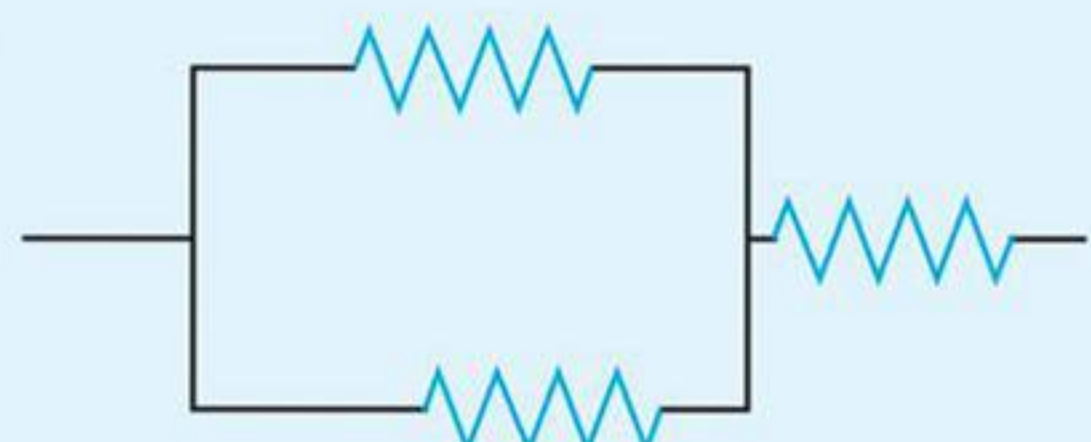
- to the intensities of lamps A and B ,
- to the intensity of lamp C ,
- to the current in the circuit, and
- to the voltage drop across the three lamps?

Does the power dissipated in the circuit increase, decrease, or remain the same?

- An electric motor operating on a 50 V dc supply draws a current of 12 A . If the efficiency of the motor is 30% , estimate the resistance of the windings of the motor.
- A fuse with a circular cross-sectional radius of 0.15 mm blows at 15 A . What should be the radius of the cross section of a fuse made of the same material that blows at 30 A ?
- A motor operating on 120 V draws a current of 2 A . If the heat is developed in the motor at the rate of 9 cal s^{-1} , what is its efficiency?
- The walls of a closed cubical box of edge 40 cm are made of a material of thickness 1 mm and thermal conductivity $4 \times 10^{-4}\text{ cal s}^{-1}\text{ cm}^{-1}\text{ }^\circ\text{C}^{-1}$. The interior of the box is maintained at 100°C above the outside temperature by a heater placed inside the box and connected across 400 V dc. Calculate the resistance of the heater.
- Two tungsten lamps with resistances R_1 and R_2 , respectively, are connected first in parallel and then in series in a lighting circuit of negligible internal resistance. Given that $R_1 > R_2$.
 - Which lamp will glow more brightly when they are connected in parallel?
 - If the lamp of resistance R_1 now burns out, how will the net illumination produced change?
 - Which lamp will glow more brightly when they are connected in series?
 - If the lamp of resistance R_2 now burns out and lamp R_1 alone is plugged in, will the net illumination increase or decrease?
- n identical bulbs are connected in series and illuminated by a power supply. One of the bulbs gets fused. The fused bulb is removed, and the remaining bulbs are again illuminated by the same power supply. Find the fractional change in the illumination of (a) all the bulbs and (b) one bulb.
- An electric motor is designed to work at 100 V and draws a current of 6 A . The output power supplied by the motor is 150 W , and the remaining goes to heat. What is the resistance of the windings of the motor and its percentage efficiency?
- A voltage stabilizer restricts the voltage output to $220\text{ V} \pm 1\%$. If the electric bulb rated 220 V , 100 W is connected to it, what will be the minimum and maximum power consumed by it?

CONCEPT APPLICATION EXERCISE 7.2

- Each of the three resistors in figure has a resistance of $2\ \Omega$ and can dissipate a maximum of 18 W without becoming excessively heated. Find the maximum power the circuit can dissipate.



11. The efficiency of a cell when connected to a resistance R is 60%. What will be its efficiency if the external resistance is increased by six times?

ANSWERS

1. 27 W

2. (a) Intensities of bulbs A and B will increase and become 2.25 of their initial values (b) Zero (c) The current in the circuit will increase and will become 1.5 times its initial value (d) The voltage drop across A and B will increase while across C it will decrease and become zero. Power dissipated in the circuit increases and becomes 1.5 times its initial value.

3. 2.9Ω 4. 0.238 mm 5. 84.25% 6. 9.92Ω

7. (a) Bulb having lower resistance will shine brightly
(b) Net illumination will decrease (c) The lamp having higher resistance will glow more brightly (d) Illumination gets increased

8. (a) $\frac{1}{n-1}$ (b) $\frac{2n-1}{(n-1)^2}$

9. 12.5Ω , 25%

10. $P_{\max} = 102 \text{ W}$, $P_{\min} = 98 \text{ W}$

11. 90%

Chapter 7

Concept Application Exercises

Exercise 7.1

- Resistance of a bulb is given by $R = V^2/P$. Now, 100 W bulb will have less resistance. So we are connecting less resistance in series with the heater. It means potential difference across the heater will increase. So the heater will give more heat.
- We know $R = V^2/P$. Therefore, resistance of the first bulb is $R_1 = V^2/P_1$. And resistance of the second bulb is $R_2 = V^2/P_2$. In series, same current will pass through each bulb. Therefore, power developed across the first bulb is $P_1' = I^2 (V^2/P_1)$ and that across the second bulb is $P_2' = I^2 (V^2/P_2)$.

$$\frac{\text{Power in bulb 1}}{\text{Power in bulb 2}} = \frac{P_2}{P_1} < 1 \text{ (as } P_2 < P_1 \text{)}$$

Hence, the bulb rated 220 V and 40 W will glow more.

- Here, $P = 30 \text{ W}$, $V = 6 \text{ V}$. Therefore, resistance of the bulb is

$$R_1 = V^2/P = (6)^2/30 = 1.2 \Omega$$

Current capacity of the bulb

$$I = \frac{P}{V} = \frac{30}{6} = 5 \text{ A}$$

Supply voltage, $V' = 120 \text{ V}$

Let R_2 be the resistance used in series with the bulb to have a current of 5 A in the circuit. Total resistance is

$$R_2 + R_1 = R_2 + 1.2$$

Therefore, current $I = V'/(R_2 + 1.2)$

$$\text{or } 5 = \frac{120}{R_2 + 1.2} \text{ or } R_2 = \frac{120}{5} - 1.2 = 22.8 \Omega \text{ in series}$$

Alternatively

Potential difference across the bulb should not exceed 6 V.

$$\text{For this } \frac{6}{114} = \frac{R_1}{R_2} \text{ or } R_2 = \frac{114}{6} R_1 = \frac{114}{6} \times 1.2 = 22.8 \Omega$$

- Since $m = \rho Al$, the lengths and hence the resistances of the wires are in the ratio 1:2. Also when they are connected in series, heat produced will be in the ratio 1:2, because $H = I^2 R$.

$$5. P = \frac{V^2}{R} = \frac{V^2 A}{\rho l}$$

To boil water in 10 min, power should be increased. To increase power, length should be decreased.

- Voltage drop across line is 10 V, and current is $10/2 = 5 \text{ A}$. Current drawn by one lamp is $100/230 \text{ A}$. So number of lamps is $5 \times 230/100 = 11.5$. Hence, maximum number of lamps is 11.
- Total wattage consumed (units) is

$$\frac{7 \times 40 \times 6}{1000} + \frac{2 \times 60 \times 6}{1000} + \frac{5 \times (220 \times 0.4) \times 1}{1000} + \frac{(220)^2}{48.4} \times \frac{10}{1000}$$

$$= 12.84 \text{ per day}$$

Number of units consumed in one month (Jan) is $12.84 \times 31 = 398.04$. Total bill is $2 \times 398.04 = ₹ 796.08$.

$$8. (a) R_1 = \frac{200^2}{300}, R_2 = \frac{200^2}{600}$$

The resistance of the first bulb is more. Hence, in series, the first bulb will produce more illumination.

$$(b) P = \frac{P_1 P_2}{P_1 + P_2} = \frac{300 \times 600}{300 + 600} = 200 \text{ W}$$

$$(c) P = P_1 + P_2 = 300 + 600 = 900 \text{ W}$$

- 25 W bulb will glow brighter. This is because $P = I^2 R$ where I is the current flowing and R is the resistance of the appliance. When I is same, $P \propto R$. The resistance of 25 W bulb is more and therefore P is more.

Exercise 7.2

$$1. P_{\max} = (i_{\max})^2 R \text{ or } 18 = i_{\max}^2 \times 2 \text{ or } i_{\max} = 3 \text{ A}$$

Here $R_{\text{eq}} = 3 \Omega$.

So the maximum power of circuit is $i_{\max}^2 R_{\text{eq}} = 27 \text{ W}$

- As the three bulbs are in series and identical, initially when switch S is open,

$$V_A = V_B = V_C = \frac{1}{3} V$$

$$\text{Also } P_A = P_B = P_C = \frac{(V/3)^2}{R} = \frac{V^2}{9R} = P$$

Now, the switch S is closed,

(a) The bulb C is short-circuited and hence potential difference across it is $V'_C = 0$ and so $V'_A = V'_B = V/2$ with $V'_C = 0$. And hence,

$$P'_A = P'_B = \frac{(V/2)^2}{R} = \frac{V^2}{4R}$$

$$\text{i.e., } P'_A = P'_B = \frac{9}{4} P,$$

i.e., intensities of bulbs A and B will increase and become 2.25 of their initial values.

(b) As $V'_C = 0$, no current will pass through bulb C , so it will give no light, i.e., $P'_C = 0$.

(c) As R_T changes from $3R$ to $2R$, so the current in the circuit will change from

$$I = \frac{V}{3R} \text{ to } I' = \frac{V}{2R}, \text{ i.e., } I' = \frac{3}{2} I$$

Thus, current in the circuit will increase and will become 1.5 times its initial value.

(d) As explained earlier, initially the voltage V divides equally across A , B , and C , i.e., $V_A = V_B = V_C = V/3$. Now when the switch S is closed, bulb C is short-circuited, i.e., $V'_C = 0$, so voltage V now will divide equally across A and B , i.e., $V'_A = V'_B = V/2$. So voltage across bulbs A and B will change from $V/3$ to $V/2$, while across C from $V/3$ to zero, i.e., voltage drop across A and B will increase while across C it will decrease and become zero. Further as R_T changes from $3R$ to $2R$, so power consumed in the circuit changes from

$$P_T = \frac{V^2}{3R} \text{ to } P'_T = \frac{V^2}{2R}, \text{ i.e., } P'_T = \frac{3}{2} P_T$$

Thus, power dissipated in the circuit increases and becomes 1.5 times its initial value.

- Power drawn by motor is $P = VI = 50 \times 12 = 600 \text{ W}$

$$\text{Power loss} = \frac{70}{100} \times 600 = 420 \text{ W}$$

This power is lost in the form of heat in the resistance.

$$I^2 R = 420 \text{ or } (12)^2 R = 420 \text{ or } R = 2.9 \Omega$$

4. Heat produced = heat loss through surface of wire

$$I^2 R = H 2\pi r l$$

where H is the heat loss per unit surface area of the wire.

$$I^2 \frac{\rho l}{a} = H 2\pi r l$$

Put $a = \pi r^2$, therefore $I^2 \propto r^3$

$$\text{or } \left(\frac{15}{30}\right)^2 = \left(\frac{0.15}{r}\right)^3 \text{ or } r = 0.238 \text{ mm}$$

5. $P_{\text{in}} = 12 \times 2 = 240 \text{ W}$, heat loss = $9 \text{ cal s}^{-1} = 9 \times 4.20 = 378 \text{ J s}^{-1}$

$$P_{\text{out}} = 240 - 37.8 = 202.2 \text{ W}$$

$$\text{Efficiency } \eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100 = 84.25\%$$

6. Heat produced = heat loss or $\frac{V^2}{R} = \frac{KA\Delta T}{l}$

Here $V = 400 \text{ V}$, $K = 4 \times 10^{-4} \text{ cal/cm s}^\circ\text{C} = 16.8 \times 10^{-4} \text{ J/cm s}^\circ\text{C}$

$$\Delta T = 100, l = 1 \text{ mm} = 0.1 \text{ cm}, A = 6(40)^2 \text{ cm}^2$$

$$\frac{400^2}{R} = \frac{16.8 \times 10^{-4} \times 6 \times (40)^2 \times 100}{0.1} \text{ or } R = 9.92 \Omega$$

7. (a) When the lamps are connected in parallel, potential difference V across each lamp will be the same and will be equal to potential difference necessary for full brightness of each bulb. As $P = V^2/R$, illumination produced by the second bulb will be higher than that produced by the first bulb, i.e., the bulb having lower resistance will shine more brightly.

(b) When R_1 burns out, power is dissipated in R_2 only. Because internal resistance is quite low in lighting circuit, potential difference is still equal to V , hence power is dissipated in the second lamp, i.e.,

$$\frac{V^2}{R_2} < \left(\frac{V^2}{R_1} + \frac{V^2}{R_2} \right), \text{ net power consumed initially.}$$

In other words, net illumination will now decrease.

(c) When the two lamps are connected in series, the potential difference across each lamp will be different, but current I flowing through each lamp will be the same, as $P = I^2 R$.

Hence, illumination produced by the first lamp will be higher as compared to that produced by the second lamp, i.e., the lamp having higher resistance will glow more brightly.

(d) When lamp of resistance R_2 burns out and only lamp of resistance R_1 is connected in the circuit, then current flowing through the circuit will change. Let new current be i' . Because potential difference still remains the same (due to low internal resistance),

$$i' R_1 = i(R_1 + R_2) \text{ or } i' = [i(R_1 + R_2)] / R_1$$

If P' is the power consumed, then

$$P' = i'^2 R_1 = [i^2 (R_1 + R_2)(R_1 + R_2)] / R_1$$

When both the lamps were present, then total power consumed was given by

$$P_s = P_1 + P_2 = i^2 (R_1 + R_2), \text{ i.e., illumination gets increased when one bulb is used.}$$

8. Let the voltage supply be V . Let resistance of each bulb be R . Total resistance of circuit is nR . Power illuminated by all bulbs,

$$P_1 = \frac{V^2}{nR}$$

Power illuminated by one bulb,

$$P_2 = \frac{P_1}{n} = \frac{V^2}{n^2 R}$$

After one bulb is fused, the powers are

$$P'_1 = \frac{V^2}{(n-1)R}, P'_2 = \frac{V^2}{(n-1)^2 R}$$

- (a) Fractional change in the illumination of all the bulbs is

$$f_1 = \frac{P'_1 - P_1}{P_1} = \frac{P'_1}{P_1} - 1 = \frac{n}{n-1} - 1 = \frac{1}{n-1}$$

- (b) Fractional change in the illumination of one bulb is

$$f_2 = \frac{P'_2 - P_2}{P_2} = \frac{n^2}{(n-1)^2} - 1 = \frac{2n-1}{(n-1)^2}$$

9. Power input = $P_i = VI = 100 \times 6 = 600 \text{ W}$

Power output = 150 W

$$\text{Efficiency: } \eta = \frac{150}{600} \times 100 = 25\%$$

Heat produced is $600 - 150 = 450 \text{ W}$

or $I^2 R = 450$ where R is the resistance of windings

$$\text{or } R = \frac{450}{6^2} = \frac{450}{36} = 12.5 \Omega$$

10. Let stabilizer give voltage V , then power produced by the bulb is

$$P = \left(\frac{V}{220} \right)^2 100 \text{ or } \frac{dP}{P} = 2 \frac{dV}{V}$$

$$\text{If } \frac{dV}{V} = \pm 1\%, \text{ then } \frac{dP}{P} = \pm 2\%$$

Hence, the maximum power produced is

$$P_{\text{max}} = 100\% + 2\% = 102 \text{ W}$$

Minimum power produced is

$$P_{\text{min}} = 100\% - 2\% = 98 \text{ W}$$

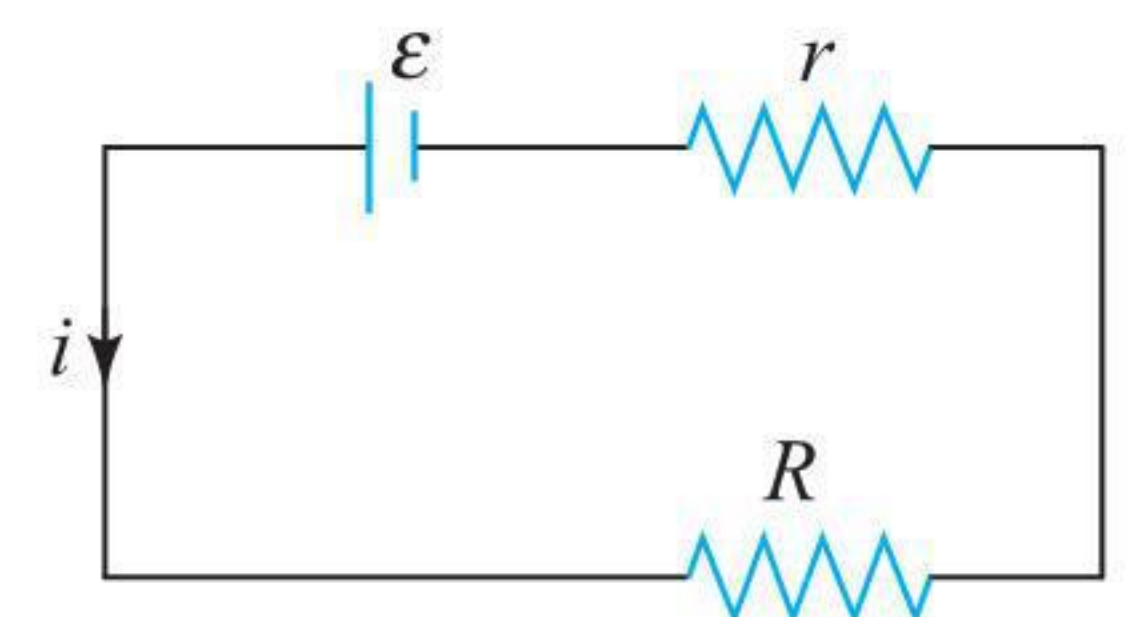
11. Efficiency = $\eta = \frac{\text{Output power}}{\text{Input power}}$

$$\eta = \frac{i^2 R}{\varepsilon i} \text{ where } i = \frac{\varepsilon}{R+r}$$

$$\eta = \frac{R}{R+r} \text{ or } 0.6 = \frac{R}{R+r} \text{ or } 2R = 3r$$

Therefore, new efficiency

$$\eta_1 = \frac{6R}{6R+r} = 0.9 = 90\%$$



Exercises

Single Correct Answer Type

$$1. (4) R_{200} = \frac{200 \times 200}{100} = 400 \Omega$$

$$\text{So } 400 = R_0 [1 + 0.005 \times 2000]$$

$$\text{or } R_0 = \frac{400}{11} \Omega \approx 36 \Omega$$

Hence, current

$$I = \frac{200}{R_0} = 5.5 \text{ A}$$

$$2. (1) \text{ Power } P = I^2 R = \left(\frac{V}{R+3r} \right)^2 R \text{ or } P \propto V^2$$

$$\text{Power ratio} = \left(\frac{V_1}{V_2} \right)^2 = \left(\frac{4.5}{1.5} \right)^2 = 3^2 = 9$$

3. (1) Maximum current flows through bulb 1.

4. (2) When switch S_2 is closed, the total current shall flow through the connecting wire only, which is supposed to have zero resistance.
5. (4) Suppose V is the voltage of the supply and R is the resistance of each bulb. Now, $R_p = R/3$ and current in ammeter is $I = V/R_p = 3V/R$, provided all three bulbs are working properly. If one bulb has broken down, then

$$R_p = \frac{R}{2} \quad \text{and} \quad I = 2\frac{V}{R}$$

Therefore, current decreases and since current through each bulb is V/R , the same as before, brightness of bulbs is not affected.

6. (1) Consider two extreme cases: (i) When the resistance of the rheostat is zero, the current through Q is zero since Q is short-circuited. The circuit is then essentially a battery in series with lamp P . (ii) When the resistance of the rheostat is very large, almost no current flows through it. So the currents through P and Q are almost equal. The circuit is essentially a battery in series with lamps P and Q .

7. (4) Let current I flow through the circuit.

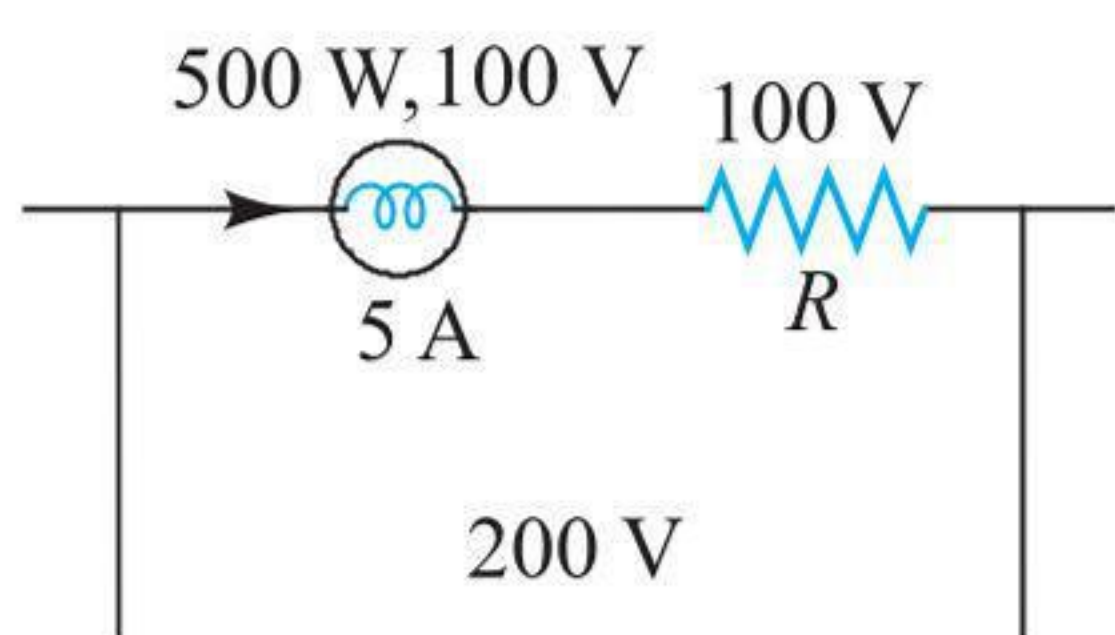
Energy dissipated in load is $I^2 R$.
Energy dissipated in the complete circuit is $I^2 (r + R)$.

Therefore, the efficiency is

$$\frac{I^2 R}{I^2 (R + r)} = \frac{R}{R + r}$$

8. (2) Let both the batteries be connected in series. Power output is maximum when external resistance is 2Ω . Current in the circuit is $4 \text{ V} / 4 \Omega = 1 \text{ A}$ and power in the external circuit is $(1)^2 \times 2 = 2 \text{ W}$.

9. (2) $P = VI$ or $I = \frac{P}{V}$ or $I = \frac{500 \text{ W}}{100 \text{ V}} = 5 \text{ A}$



$$\text{Now, } 5R = 100 \quad \text{or} \quad R = 20 \Omega$$

10. (2) $P = \frac{V^2}{l} \rho \frac{1}{a}$ or $P \propto \frac{1}{l}$, $P' \propto \frac{1}{l - \frac{10}{100}l}$
- $$\frac{P'}{P} = \frac{10}{9} \quad \text{or} \quad \left(\frac{P'}{P} - 1 \right) \times 100 = \left(\frac{10}{9} - 1 \right) \times 100$$
- $$\frac{P' - P}{P} \times 100 = \frac{100}{9} \approx 11$$

11. (2) Heat developed is

$$H = \frac{V^2}{R} t$$

Heat developed will be doubled when R is halved. Further, $R = \rho l / (\pi r^2)$ and $H = V^2 \pi r^2 t / \rho l$. So heat produced will be doubled when both the length and radius of the wire are doubled.

12. (2) Heat produced $\propto I^2$, initial heat produced $= k (20)^2$, final heat produced $= 2k (20)^2$. If final current is I' , then

$$kI'^2 = 2 \times k \times (20)^2 \quad \text{or} \quad I' = 20\sqrt{2} \text{ A}$$

13. (2) Power $= \frac{V^2}{R}$

$$\text{or} \quad V = \sqrt{PR} = \sqrt{2 \times 10} = \sqrt{20} = 2\sqrt{5} \text{ V}$$

Clearly, voltage across single resistor of 10Ω cannot exceed $2\sqrt{5} \text{ V}$. Note that the resistance of parallel combination is half of 10Ω . Thus, the maximum possible voltage between A and B is $3\sqrt{5} \text{ V}$.

14. (4) Resistance of bulb is

$$\frac{1.5 \times 1.5}{4.5} \Omega = 0.5 \Omega$$

Resistance of parallel combination,

$$R = \frac{1 \times \frac{1}{2}}{1 + \frac{1}{2}} \Omega = \frac{1}{3} \Omega$$

$$\text{Now, } r = \frac{E - V}{V} R \quad \text{or} \quad \frac{8}{3} = \frac{E - 1.5}{1.5} \times \frac{1}{3} \quad \text{or} \quad E = 13.5 \text{ V}$$

15. (3) Power of the heater is $P = 1000 \text{ W}$. Potential difference is $V = 100 \text{ V}$. Therefore, resistance is

$$R_1 = \frac{V^2}{P} = \frac{100 \times 100}{1000} = 10 \Omega$$

Now resistance of the circuit is

$$R_2 = 10 + \frac{10R}{10 + R} = \frac{100 + 20R}{10 + R}$$

Therefore, power is

$$\frac{V^2}{R_2} = 625 \text{ W}$$

$$\text{or} \quad R_2 = \frac{V^2}{625} \quad \text{or} \quad \frac{100 + 20R}{10 + R} = \frac{625}{100 \times 100} \Rightarrow R = 15 \Omega$$

16. (4) $R_{60} = \frac{120 \times 120}{60} \Omega = 240 \Omega$

$$\text{Current} = \frac{120}{240 + 6} \text{ A} = \frac{120}{246} \text{ A}$$

$$\text{Voltage across bulb} = \frac{120}{246} \times 240 \text{ V} = 117.1 \text{ V}$$

$$R_{240} = \frac{120 \times 120}{240} \Omega = 60 \Omega$$

Resistance of parallel combination is

$$\frac{60 \times 240}{60 + 240} \Omega = 48 \Omega$$

Total resistance is $(48 + 6) \Omega = 54 \Omega$.

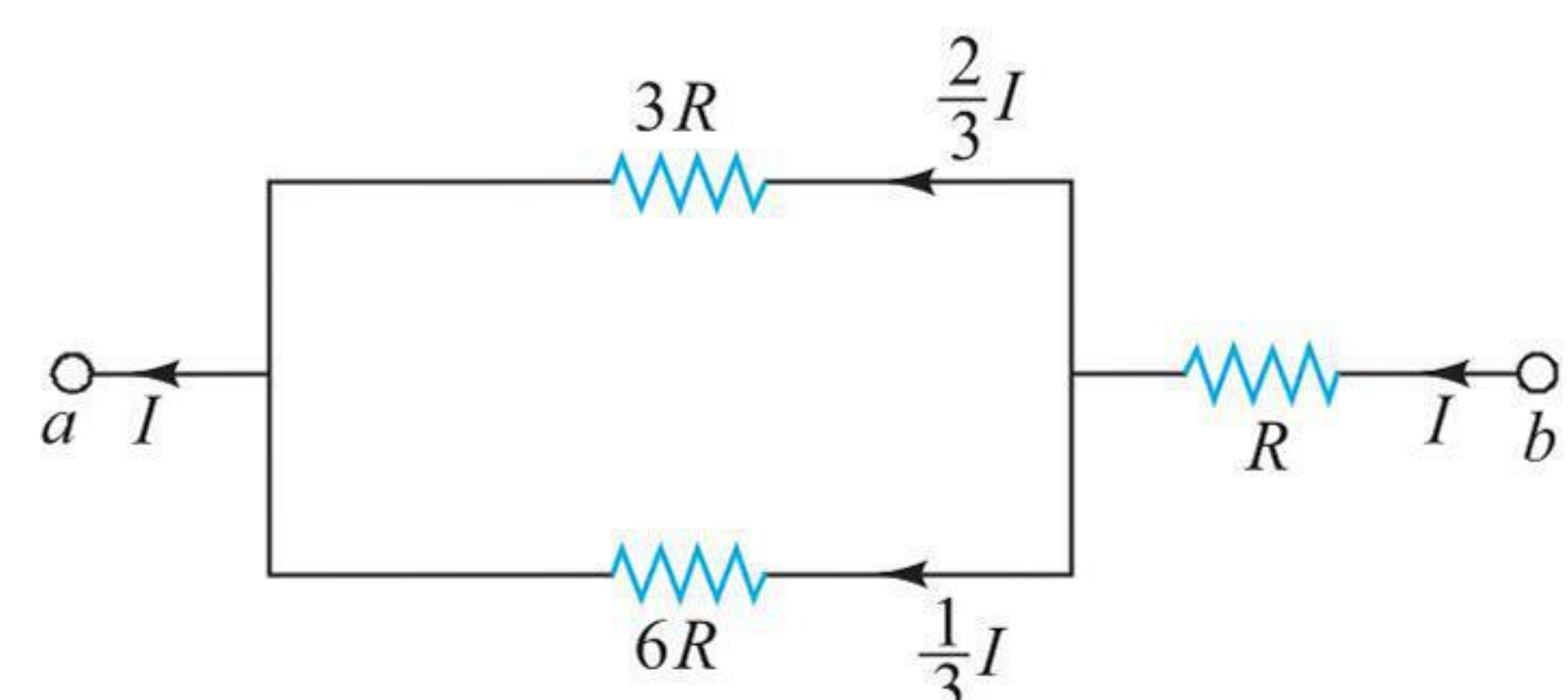
Current I is $120/54 \text{ A}$. Voltage across parallel combination is

$$\frac{120}{54} \times 48 \text{ V} = 106.7 \text{ V}$$

Change in voltage is $(117.1 - 106.7) \text{ V} = 10.4 \text{ V}$.

17. (3) Let current flow from b to a as shown (figure).

$$\text{Ratio is } \left(\frac{2}{3} I \right)^2 3R : \left(\frac{1}{3} I \right)^2 6R : I^2 R \quad \text{or} \quad \frac{4}{3} : \frac{2}{3} : 1 \quad \text{or} \quad 4 : 2 : 3$$



18. (1) Let resistors P , Q , and R have resistance r . The effective resistance across the source is

$$R_{\text{eff}} = r + r \parallel r = r + \frac{(r)(r)}{r+r} = r + \frac{r}{2} = \frac{3r}{2}$$

Current drawn from source is

$$I_s^2 R_{\text{eff}} = 12$$

$$\text{or } I_s = \sqrt{\frac{12}{R_{\text{eff}}}} = \sqrt{\frac{8}{r}} \text{ A}$$

Since Q and R have equal resistance r , each draws a current of I , which is given by

$$I = \frac{1}{2} I_s = \sqrt{\frac{2}{r}} \text{ A}$$

Heat dissipation in R can now be determined and is given by

$$I^2 r = \left(\frac{2}{r}\right) r = 2 \text{ W}$$

19. (1) Since all the three resistors are in series, the same current will flow through them. Their resistances are given by $R = V^2/P$.

$$R_{40} = \frac{240 \times 240}{40} = 1440 \Omega$$

$$R_{60} = 960 \Omega \quad \text{and} \quad R_{100} = 576 \Omega$$

Total resistance is $R = 2976 \Omega$

$$I = \frac{240}{2976} = 0.0806 \text{ A}$$

Potential difference across the 40 W bulb is

$$1440 \times 0.0806 = 116 \text{ V}$$

20. (2) Let current in 5Ω resistor be I_1 and in 4Ω resistors be I_2 , then $I_2/I_1 = 1/2$. The heat generated in the 5Ω resistor is 10 cal s^{-1} . Given $I_1^2 \times 5 = 10$ or $(2I_2)^2 \times 5 = 10$ or $I_2^2 = 1/2$. Heat in the 4Ω resistor will be $(I_2)^2 \times 4 = 2 \text{ cal s}^{-1}$.

21. (2) For the maximum power, external resistance is equal to internal resistance. Therefore, $2R = 4$ or $R = 2 \Omega$.

22. (1) Net resistance of circuit, $R = \frac{10 \times 5}{10 + 5} + \frac{10 \times 5}{10 + 5} = \frac{20}{3} \Omega$

Heat generated in circuit per minute is

$$Q = I^2 R t = (10)^2 \times \frac{20}{3} \times (10 \times 60)$$

$$= 4 \times 10^5 \text{ J} = \frac{4 \times 10^5}{4.2 \times 10^3} \text{ kcal}$$

$$m = \frac{Q}{L} = \frac{4 \times 10^5}{4.2 \times 10^3 \times 80} = 1.19 \text{ kg}$$

23. (1) $P_I = I^2(3R)$, $P_{II} = I^2\left(\frac{2R}{3}\right)$, $P_{III} = I^2\left(\frac{R}{3}\right)$

$$P_{IV} = I^2\left(\frac{3R}{2}\right)$$

$$\therefore III < II < IV < I$$

24. (2) The heat supplied under these conditions is the change in internal energy $Q = \Delta U$. The heat supplied is

$$Q = i^2 R t = 1 \times 1 \times 100 \times 5 \times 60 = 30000 \text{ J} = 30 \text{ kJ}.$$

25. (1) All resistances are in parallel.

So for parallel combination

$$H = \frac{V^2}{R} t \text{ or } H \propto \frac{1}{R}$$

26. (4) Let the resistance of the lamp filament be R . Then $100 = (220)^2/R$. When the voltage drops, expected power is $P = (220 \times 0.9)^2/R'$. Here, R' will be less than R , because now

the rise in temperature will be less. Therefore, P is more than $(220 \times 0.9)^2/R = 81 \text{ W}$. But it will not be 90% of the earlier value, because fall in temperature is small. Hence, option (4) is correct.

27. (2) In the first case,

$$\frac{(3E)^2}{R} t = (m) s \Delta T \quad \dots(i)$$

When the length of the wire is doubled, resistance and mass both are doubled. Therefore, in the second case,

$$\frac{(NE)^2}{2R} t = (2m) s \Delta T \quad \dots(ii)$$

Dividing Eq. (ii) by Eq. (i), we get $N = 6$.

28. (4) L is the latent heat of vaporization of water, the heat required for producing 1 g of steam. So

$$L = 540 \text{ cal} = 540 \times 4.2 = 2268 \text{ J}$$

Energy supplied is 1080 Js^{-1} . Time required to boil 100 g of water is

$$t = 540 \times 4.2 \times 100/1080 = 210 \text{ s}$$

29. (3) Given $P_A = 60 \text{ W}$, $P_B = 100 \text{ W}$. We know that current flowing through the bulb is $I = P/V$. We also know that as both the bulbs are connected in parallel, potential difference (V) across both the bulbs is the same. Thus, $I \propto P$. Since the power of bulb B is greater than that of bulb A , bulb B draws more current than bulb A .

30. (1) As both bulbs are in series, current through them will be the same. But resistance of 25 W bulb is more. Hence, from the relation $P = I^2 R$, the 25 W bulb will glow more brightly.

31. (3) Let the resistance of the two heaters be denoted by R_1 and R_2 , respectively. Then

$$R_1 = \left(\frac{V^2}{P_1}\right) \text{ and } R_2 = \left(\frac{V^2}{P_2}\right)$$

If the resistance of the series combination is denoted by R_s and the corresponding power by P_s , then $R_s = R_1 + R_2$. So

$$\frac{V^2}{P_s} = \frac{V^2}{P_1} + \frac{V^2}{P_2}$$

$$\text{or } P_s = \frac{P_1 P_2}{P_1 + P_2} = \frac{1000 \times 1000}{2000} = 500 \text{ W}$$

32. (2) Let R be the resistance of each lamp. If E is the applied emf, then the current in the circuit I_1 is given by

$$I_1 = \frac{E}{R + (R/2)} = 2E/3R$$

Current flowing through L_2 or L_3 is

$$\frac{1}{2} \left[\frac{2E}{3R} \right] = \frac{E}{3R}$$

When L_3 is fused, the whole current flows through L_1 and L_2 . Thus $I_2 = E/2R$. So current through L_1 decreases and the current through L_2 increases.

33. (3) The rate of heat developed is given by

$$\frac{V^2}{R} = \frac{(210)^2}{20} = 2205 \text{ Js}^{-1} = \frac{2205}{4.2} \text{ cal s}^{-1}$$

Let m be the amount of ice melting per sec and, then

$$m \times 80 = \frac{2205}{4.2} \quad \text{or} \quad m = 6.56 \text{ gs}^{-1}$$

34. (2) Current required by each bulb is

$$I = \frac{P}{V} = \frac{100}{220} \text{ A}$$

If n bulbs are joined in parallel, then

$$nI = I_{\text{fuse}} \quad \text{or} \quad n \times \frac{100}{220} = 10 \quad \text{or} \quad n = 22$$

35. (2) If N is the initial number of turns in the coil and r is the radius of coil, then its resistance is

$$R = \rho \frac{L}{A} = \rho \frac{N 2\pi r}{A} \quad \text{or} \quad \frac{V^2 t A}{4.2 \rho N 2\pi r} = Q = msd\theta$$

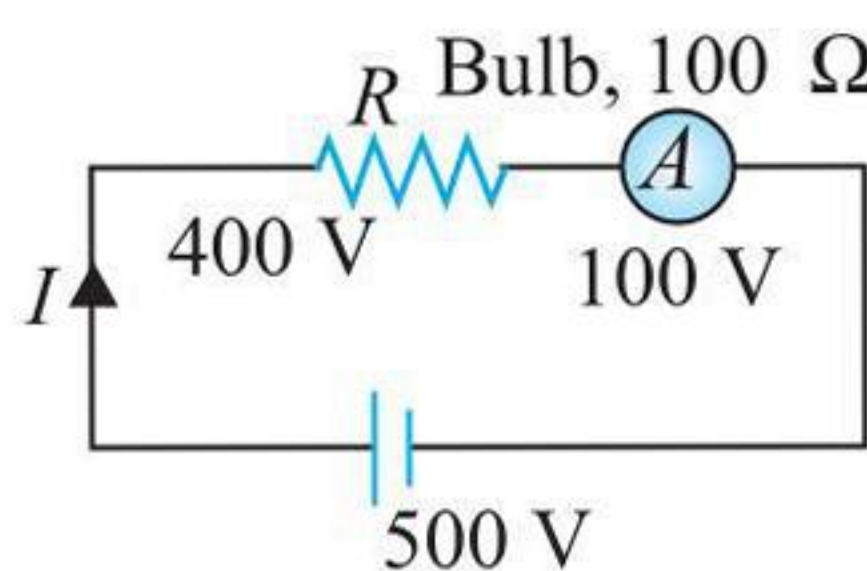
or $\frac{t}{n} = \frac{ms d\theta \times 4.2 \rho 2\pi r}{V^2 A} = \text{constant}$

or $\frac{t_1 / N_1}{t_2 / N_2} = 1$ or $t_2 = \frac{N_2}{N_1} t_1 = \left(\frac{9}{10} \frac{N_1}{N_1} \right) t_1 = \frac{9}{10} \times 16 \text{ min}$

36. (3) Resistance of bulb is $R_b = \frac{V^2}{P} = \frac{(100)^2}{100} = 100 \Omega$

If bulb consumes 100 W, then potential difference across the bulb should be 100 V. So $I = 100/100 = 1$ A. And potential difference across R is $500 - 100 = 400$ V. So

So $R = \frac{400}{I} = \frac{400}{1} = 400 \Omega$



37. (4) $P = VI$ or $50 = 5 \times I$ or $I = 10$ A

Power lost in cable $= I^2 R = 10 \times 10 \times 0.2 = 2$ W

Power supplied to the tape recorder $= 50 \text{ W} - 2 \text{ W} = 48 \text{ W}$

38. (3) In this option, emf is least and resistance is maximum, so bulbs will be dimmest in this option.

39. (3) Initially, $R_{\text{eq}} = 5R/3$. Finally, $R_{\text{eq}} = 3R/2$.

Equivalent resistance decreases, so current increases in circuit and in 1 also. Hence, brightness of 1 increases. It means pd across 1 increases, so across 2 pd decreases, hence brightness of 2 decreases.

Initially, pd across 4 is $V_{4i} = \frac{1}{2} \left[\frac{(2R/3)\epsilon}{2R/3 + R} \right] = \frac{\epsilon}{5}$

Finally, $V_{4f} = \frac{(R/2)\epsilon}{R/2 + R} = \frac{\epsilon}{3}$

Since $V_{4f} > V_{4i}$; hence, brightness of 4 increases.

40. (2) Energy stored in capacitor when it is charged up to 2 V is

$$\frac{1}{2} \times 10 \times 2^2 = 20 \mu\text{J} = u_1 \text{ (suppose)}$$

Energy stored in capacitor when it is charged up to 4 V is

$$\frac{1}{2} \times 10 \times 4^2 = 80 \mu\text{J} = u_2 \text{ (suppose)}$$

Increase in charge $= 40 - 20 = 20 \mu\text{C}$

Energy drawn from cell $= 20 \times 4 = 80 \mu\text{J} = u$ (suppose)

Heat produced $= u_1 + u - u_2$
 $= 20 + 80 - 80 = 20 \mu\text{J}$

41. (3) $i = \frac{dq}{dt}$ = slope of $q-t$ graph

$$= -5 \text{ (which is constant)}$$

Amount of heat generated in time $t = i^2 R T \propto t$

42. (2) Circuit is forming a Wheatstone bridge, so $R = 2R$. For the maximum power transfer, $2R = r$.

43. (2) $H = \int_0^4 \frac{E^2}{R} dt = \int_0^4 \frac{(6t)^2}{12} dt = 64 \text{ J}$

44. (1) Initial charge on capacitor: $Q_1 = 2 \times 100 = 200 \mu\text{C}$

Final charge: $Q_2 = 2 \times 300 = 600 \mu\text{C}$

$$W_b = 300 [Q_2 - Q_1] = 12,0000 \mu\text{J}$$

$$\Delta U = \frac{1}{2} C [300^2 - 100^2] = 80000 \mu\text{J}$$

$$\text{Heat} = W_b - \Delta U = 40000 \mu\text{J} = 0.04 \text{ J}$$

45. (1) Time period: $T_0 = 2T$, $\omega = \frac{2\pi}{T_0} = \frac{\pi}{T}$

$$I = I_0 \sin \omega t = I_0 \sin \left(\frac{\pi}{T} t \right)$$

$$Q = \int_0^T I dt$$

$$\Rightarrow = I_0 \int_0^T \sin \left(\frac{\pi}{T} t \right) dt$$

$$= \frac{-I_0}{\pi/T} \left[\cos \left(\frac{\pi}{T} t \right) \right]_0^T = \frac{2TI_0}{\pi}$$

or $I_0 = Q\pi/2T$

$$\text{Heat} = \int_0^T I^2 R dt = \int_0^T I_0^2 \sin^2 \left(\frac{\pi}{T} t \right) R dt = Q^2 \pi^2 R / 8T$$

46. (1) $I = I_0 - kt$, at $t = T$, $I = 0$ or $k = I_0/T$

So $I = I_0 - I_0 t/T$, $q = \int_0^T I dt = I_0 \int_0^T \left(1 - \frac{t}{T} \right) dt$

or $q = \frac{I_0 T}{2}$ or $I_0 = \frac{2q}{T}$

$$\text{Heat} = \int_0^T I^2 R dt = I_0^2 R \int_0^T \left(1 - \frac{t}{T} \right)^2 dt = \frac{4q^2 R}{3T}$$

47. (4) $\frac{V^2}{R} t = ms\Delta T$

For first case: $\frac{(3E)^2}{(\rho L/A)} t = ms\Delta T$

For second case: $\frac{(NE)^2}{\rho 2L/A} t = 2ms\Delta T$

Dividing, we get $N = 6$.

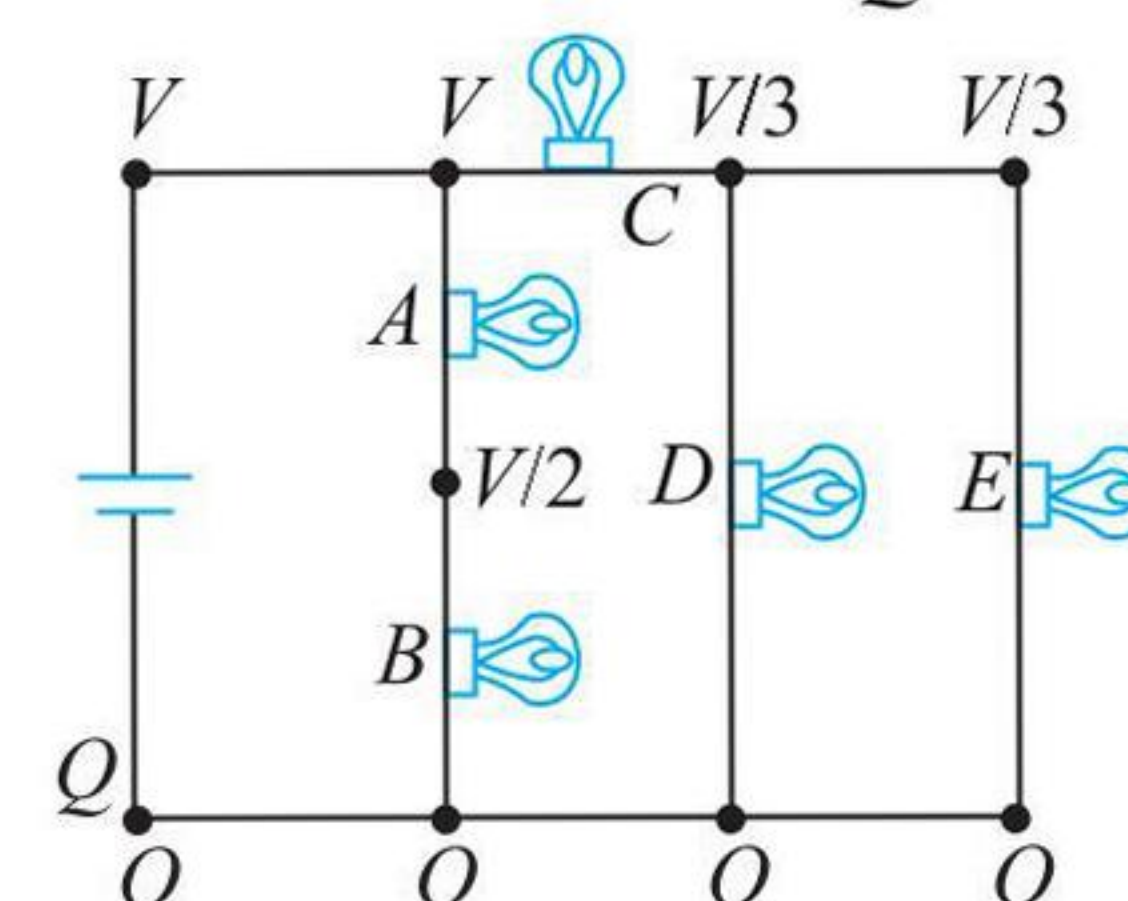
48. (4) All resistances are in parallel. The potential difference across each resistance is same.

$$\Delta H = I^2 R t = \frac{V^2 t}{R}$$

$$\Delta H \propto \frac{1}{R}$$

R_5 is the smallest and generates maximum heat.

49. (2) Let potential at P is V and that at Q is 0



Potential difference across bulb C is maximum.

50. (1) For the branch containing A , B , and C .

$$R_{\text{eq}} = \frac{3R}{2}$$

Current in A : $I_1 = \frac{E}{R_{eq}} = \frac{2E}{3R}$

Current in each of B and C :

$$I'_1 = \frac{I_1}{2} = \frac{E}{3R}$$

For the branch containing D , E , F , G , and H :

$$R_{eq} = \frac{7R}{3}$$

Current through D and H :

$$I_2 = \frac{E}{R_{eq}} = \frac{3E}{7R}$$

Current through each of E , F , and G :

$$I'_2 = \frac{I_2}{3} = \frac{E}{7R}$$

Since $I_1 > I_2 > I'_1 > I'_2$

$$A > D = H > B = C > E = F = G$$

Multiple Correct Answers Type

1. (2), (3)

$$P = \frac{V^2}{R} \text{ or } R = \frac{V^2}{P} \text{ or } R \propto V^2,$$

$$\text{i.e., } \frac{R_1}{R_2} = \left(\frac{200}{300}\right)^2 = \frac{4}{9}$$

When connected in series, potential drop is in the ratio of their resistances. So,

$$\frac{V_1}{V_2} = \frac{R_1}{R_2} = \frac{4}{9}$$

Now, $P = I^2 R$

or $P \propto R$ (in series I is the same)

$$\text{or } \frac{P_1}{P_2} = \frac{R_1}{R_2} = \frac{4}{9}$$

2. (1), (2), (3)

Current through the circuit is $I = \varepsilon/(r + R)$.

Power dissipated in external resistance is

$$P = I^2 R = \frac{\varepsilon^2}{(r + R)^2} R$$

We know that power dissipated is maximum when $r = R$.

$$P_{\max} = \frac{\varepsilon^2 r}{(2r)^2} = \frac{\varepsilon^2}{4r} = 9 \text{ W, } i = \frac{\varepsilon}{2r} = 3 \text{ A}$$

Solve to get, $\varepsilon = 6 \text{ V}$ and $r = 1 \Omega$.

3. (1), (4)

Let $V = 200 \text{ V}$ and R_1 and R_2 be the resistances of the 25 W and 100 W bulbs.

$$P_1 = 25 = V^2/R_1 \text{ or } R_1 = V^2/25$$

$$\text{and } R_2 = V^2/100$$

When the bulbs are joined in series, the current is

$$I = \frac{V}{R_1 + R_2}$$

Power in the 25 W bulb is $R_1 I^2$ and in the 100 W bulb is $R_2 I^2$.

4. (1), (4)

Resistance of upper bulb, $R_1 = 100^2/25 = 400 \Omega$

Resistance of lower bulb, $R_2 = 200^2/100 = 400 \Omega$

Equivalent resistance of the circuit, $R_{eq} = 500 \Omega$.

Heat lost per second in the circuit is

$$e^2/R_{eq} = 200^2/500 = 80 \text{ J s}^{-1}$$

Potential difference across branches AB and CD will be same. Hence, ratio of heat generated in them is

$$\frac{\text{Heat}_{AB}}{\text{Heat}_{CD}} = \frac{R_{CD}}{R_{AB}} = \frac{1000}{500} = 2:1$$

Hence, (3) is incorrect.

Since potential difference across branches AB and CD will be the same, but their resistances are different, so current in them will be different. As the resistances of the bulbs are same, so heat generated in them will be different. Hence, (2) is incorrect. Current drawn from the cell is $e/R_{eq} = 200/500 = 0.4 \text{ A}$.

5. (1), (2), (3), (4)

Potential difference across the terminals of the cell is given by $V = E - ir$. Hence, the graph between V and i will be a straight line having negative slope and positive intercept. Hence, (1) is correct.

Total electrical power generated is Ei and thermal power generated in the cell is $i^2 r$. Hence, thermal power generated in the external circuit is $P = Ei - i^2 r$. So, the graph between P and i is a parabola passing through the origin. Hence, choice (2) is correct.

When terminal potential difference across the cell is V , current flowing through the circuit will be $i = (E - V)/R$ and thermal power generated in the external circuit will be equal to

$$P = Vi = V \left(\frac{E - V}{r} \right) = \frac{EV}{r} - \frac{V^2}{r}$$

Hence, the graph between P and V will be a parabola passing through the origin and have a maximum value of P at $V = E/R$. Hence, choice (4) is correct.

At an instant, thermal power generated in the cell is equal to $i^2 r$ and total electrical power generated in the cell is equal to Ei . Hence, the fraction $\eta = i^2 r / Ei = (r/E) i$.

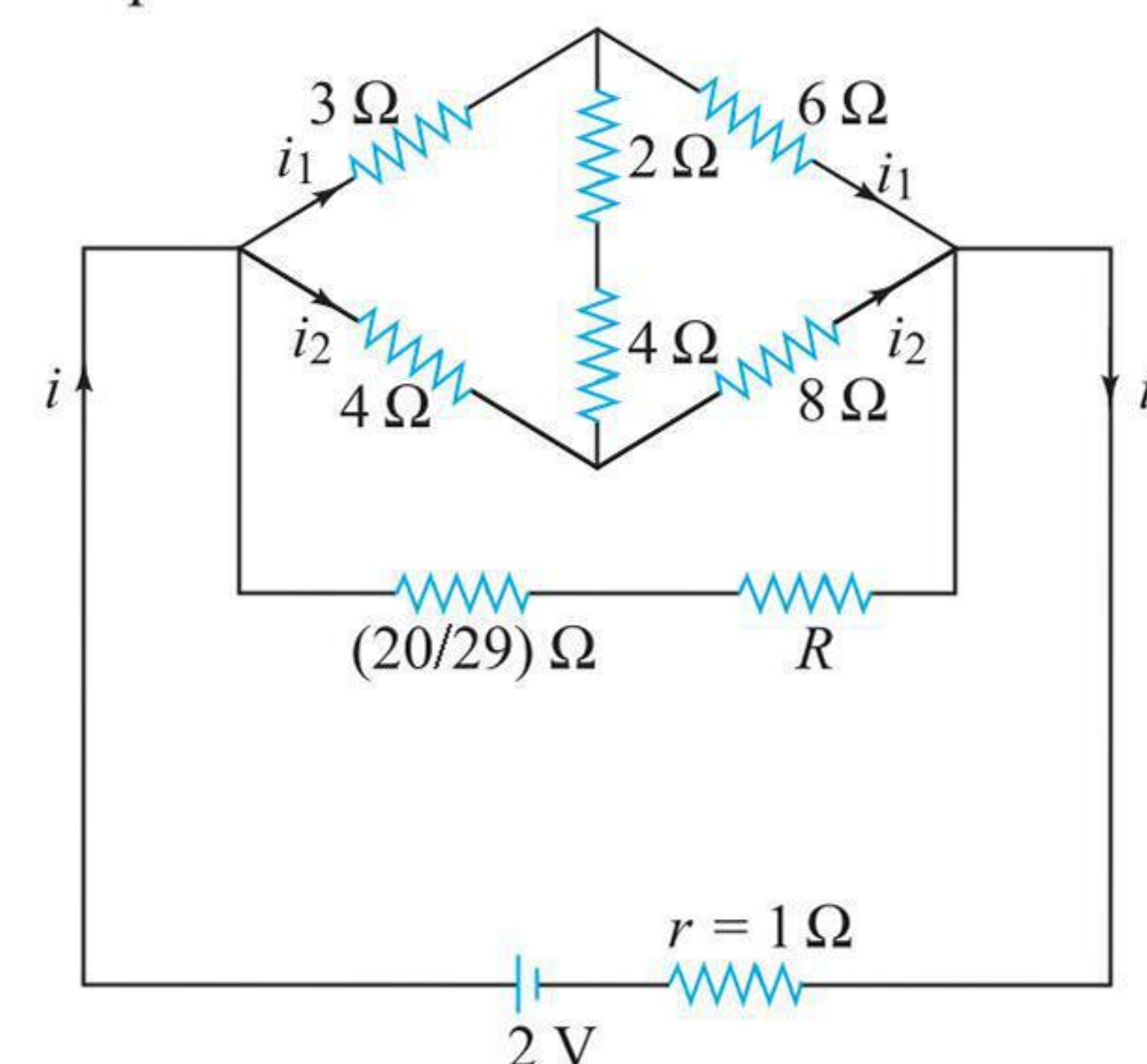
Hence, the graph between η and i will be a straight line passing through the origin and have a positive slope. Hence, (3) is correct.

6. (2), (3)

As $R \propto 1/W$ and actual power consumed $P = V^2/R$, the lesser the resistance (i.e., higher the wattage), the greater the power consumed. If the bulbs are connected in series, actual power $P = i^2 R$ or R or $P \propto R$. Hence, in series connection, low wattage bulbs glow more.

7. (1), (3)

In the simplified circuit, the circuit is a balanced Wheatstone bridge and a branch of $20/29 \Omega$ and R is parallel with this balanced bridge for maximum power.



$$r = R_{\text{external}}$$

$$1 = \frac{1}{\frac{1}{(3+6)} + \frac{1}{(20/29+R)} + \frac{1}{(4+8)}}$$

or $R = \frac{16}{25} \Omega$

Maximum power developed in the external circuit is

$$P_{\max} = i^2 R = \left(\frac{2}{1+1} \right)^2 \times 1 = 1 \text{ W}$$

Current through the upper branch

$$i_1 = i \left[\frac{\left(\frac{20}{29} + R \right) (4+8)}{9 + \left(\frac{20}{29} + R \right) + 12} \right]$$

$$i_2 = i \left[\frac{\left(\frac{20}{29} + R \right) (3+6)}{9 + \left(\frac{20}{29} + R \right) + 12} \right]$$

Therefore, i_1/i_2 is independent of R .

8. (1), (4)

H amount of heat is produced by first heater in time t_1 so we have

$$H = \frac{V^2}{R_1} t_1 \Rightarrow R_1 = \frac{V^2 t_1}{H}$$

Similarly, for second heater its resistance, $R_2 = \frac{V^2 t_2}{H}$

In series combination same heat is produced, $H = \left(\frac{V^2}{R_1 + R_2} \right) t$

$$\Rightarrow t = \frac{H(R_1 + R_2)}{V^2}$$

Substituting the values of R_1 and R_2 , we get, $t = t_1 + t_2$

In parallel combination same heat is produced in time, $H = \frac{V^2}{R_{\text{net}}} t$

$$H = V^2 t \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = V^2 t \left(\frac{H}{V^2 t_1} + \frac{H}{V^2 t_2} \right) \Rightarrow t = \frac{t_1 t_2}{t_1 + t_2}$$

9. (1), (3)

Resistances 4 and 6 Ω are short-circuited. Therefore, no current will flow through these two resistances. Current passing through the battery is $I = (20/2) = 10 \text{ A}$. This is also the current passing in wire AB from B to A . Power supplied by the battery is

$$P = EI = (20)(10) = 200 \text{ W}$$

Potential difference across the 4 Ω resistance = potential difference across the 6 Ω resistance = 0

10. (1), (2), (4)

Total charge = $\int I dt$ = Area under the curve = 10 C

$$\text{Average current} = \frac{\int I dt}{\int dt} = 5 \text{ A}$$

Total heat produced = $\int I^2 dt$

$$= \int_0^2 (-5t + 10)^2 dt = \frac{200}{3} \text{ J}$$

Maximum power = $I^2 R$ when I is maximum current
 $= 100 \times 1 = 100 \text{ W}$

11. (2), (3)

For a bulb $R = \frac{V^2}{W}$ or $R_B \leq R_A$

when switch is open $I_A = I_B$

$$P_A = R_A I_A^2$$

$$P_A = R_A I_A^2$$

$$P_B < P_A \text{ and } V_A > V_B$$

$$V_A > 12 \text{ V and } V_B < 12 \text{ V}$$

After closing the switch

$$V = V_B = 12 \text{ V or } P_B = 36 \text{ W}$$

$$P_A = 24 \text{ W}$$

12. (1), (2), (4)

Before shorting the wire at steady state the charges on each capacitors will be equal to

$$q_0 = q_1 = q_2 = \left(\frac{2C}{3} \right) V$$

After shorting the capacitor marked (2) will be useless in circuit. Hence $q'_1 = 2CV = q_f$ and $q'_2 = 0$

The charge supplied by battery after shorting

$$\Delta q = q_f - q_0 = 2CV - \frac{2CV}{3} = \frac{4CV}{3}$$

Hence work done by battery $W = \Delta q V = \frac{4CV^2}{3}$

Initial potential energy of system

$$V_0 = \frac{1}{2} \left(\frac{2}{3} C \right) V^2 = \frac{1}{3} CV^2$$

Final potential energy of system $U_f = \frac{1}{2} (2C) V^2 = CV^2$

Hence change in PE $\Delta U = U_f - U_i$

$$\text{or } \Delta U = CV^2 - \frac{1}{3} CV^2 = \frac{2}{3} CV^2$$

Work done by battery $W = \Delta U + \text{Heat}$

Heat developed $H = W - \Delta U$

$$\text{or } H = \frac{4}{3} CV^2 - \frac{2}{3} CV^2 = \frac{2}{3} CV^2$$

13. (2), (4)

Resistance absorbs energy at the rate of 2 W.

Potential difference across AB is V_{AB} . $I = 50 \text{ W}$

$$V_{AB} = 50 \text{ V}$$

Drop across resistor is 2 V, therefore emf of E is 48 V.

As AB is absorbing energy at the rate of 50 W, 48 W is being absorbed by E . Thus, E is on charging. i.e., current is entering from positive terminal of E .

14. (1), (2), (4)

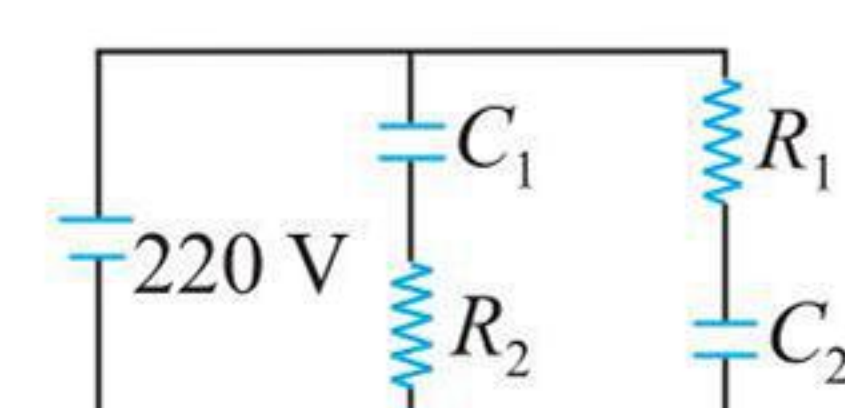
$$i = 2 \times 10^{-2} \text{ A}$$

$$P_{R_1} = i^2 R_1 = (2 \times 10^{-2})^2 \times 4 \times 10^3 = 1.6 \text{ W}$$

$$Q_{C_1} = V_{R_1} \times C_1 = 80 \times 3 \times 10^{-6} = 240 \mu\text{C}$$

$$Q_{C_2} = V_{R_2} \times C_2 = 140 \times 6 \times 10^{-6} = 840 \mu\text{C}$$

$$Q_{C_1} = EC_1 = 220 \times 3 \times 10^{-6} = 660 \mu\text{C}$$



Linked Comprehension Type

For Problems 1–3

1. (2), 2. (1), 3. (2)

$$i_2 = i_3 = I_b, \quad V_1 = (V_2 + V_3) = V, \text{ and } P_2 = P_3$$

$$\text{or } R_2 = R_3 \text{ and } V_2 = V_3 = V/2$$

$$P_1 = P_2 = P_3 = P_4 \text{ (given)}$$

$$P_1 = \frac{V^2}{36}; \quad P_3 = \frac{(V/2)^2}{R_3}$$

$$\text{As } P_1 = P_3, \text{ so } R_3 = 9 \Omega$$

$$\text{Also } R_3 = R_2 = 9 \Omega$$

$$I_a = \frac{I}{3} \text{ and } I_4 = I,$$

$$P_1 = P_4 = 4 \text{ W}$$

$$\left(\frac{I}{3}\right)^2 R_1 = (I)^2 R_4 = 4 \text{ W} \text{ or } \frac{R_1}{9} = R_4 \text{ or } R_4 = 4 \Omega$$

$$\text{Also } I = 1 \text{ A, } R_{eq} = 16 \Omega \text{ and } I = 1 \text{ A, } \varepsilon = 16 \text{ V}$$

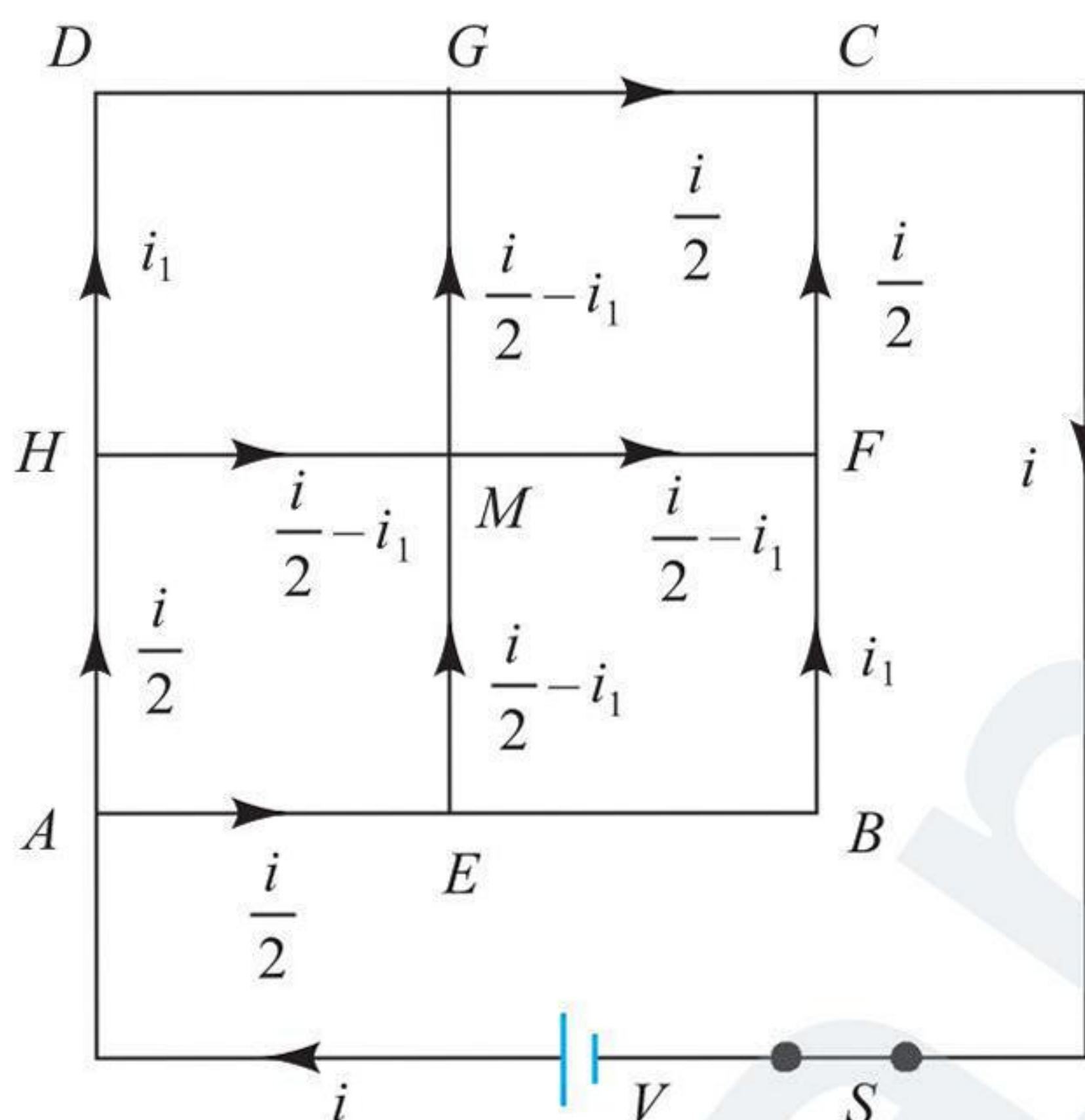
For Problems 4–6

4. (3), 5. (4), 6. (2)

In loop EBFME,

$$V_E - ri_1 - ri_1 + r\left(\frac{i}{2} - i_1\right) + r\left(\frac{i}{2} - i_1\right) = V_E$$

$$\text{or } 2i_1 = 2\left(\frac{i}{2} - i_1\right) \text{ or } i_1 = \frac{i}{4}$$



$$\text{Loop AEBCSA, } V_A - r\frac{i}{2} - ri_1 - ri_1 - r\frac{i}{2} + V = V_A$$

$$\text{or } V = ri + 2ri_1 = ri + 2r\frac{i}{4} = \frac{3ri}{2}$$

$$R_{eq} = \frac{V}{i} = \frac{3r}{2}$$

$$\frac{P_1}{P_2} = \frac{\left(\frac{i}{2}\right)^2 r}{\left(\frac{i}{2} - i_1\right)^2 r} = \frac{i^2}{(i - 2i_1)^2} = 4$$

Let l be the balancing length, then

$$\begin{aligned} kl &= V_H - V_C \\ &= \left(\frac{i}{2} - i_1\right)r + \left(\frac{i}{2} - i_1\right)r + \frac{i}{2}r \\ &= \left(\frac{3i}{2} - 2i_1\right)r = ir \end{aligned}$$

$$\text{or } l = \frac{2V}{3k}$$

For Problems 7–9

7. (4), 8. (3), 9. (1)

Initially, when all the capacitors are uncharged, they will act as conducting wires. Hence, all the resistances (except 5Ω) will be short-circuited. So no resistance occurs between A and F . Thus current through the 5Ω resistor will be $(10/5) \text{ A} = 2 \text{ A}$.

In steady state, distribution of current is shown.

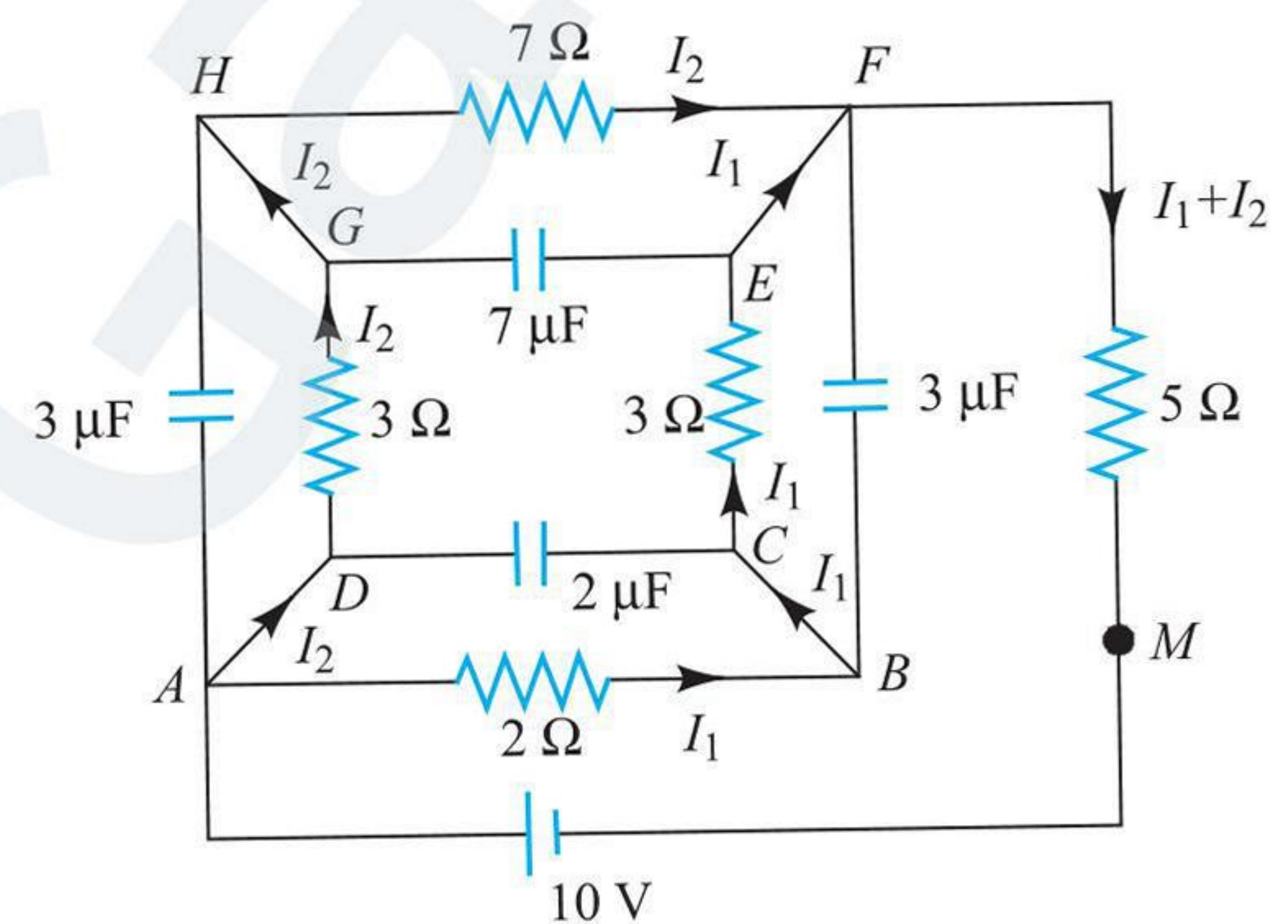
$$\text{Loop } ABCEFMA, \quad 10 = 2I_1 + 3I_1 + 5(I_1 + I_2)$$

$$\text{or } 10I_1 + 5I_2 = 10 \quad \dots(i)$$

$$\text{Loop } ADGHFMA, \quad 10 = 3I_2 + 7I_2 + 5(I_1 + I_2)$$

$$\text{or } 5I_1 + 15I_2 = 10 \quad \dots(ii)$$

$$\text{From Eqs. (i) and (ii), } I_1 = \frac{4}{5} \text{ A, } I_2 = \frac{2}{5} \text{ A}$$



Hence, current through the 5Ω resistor is $I_2 + I_2 = (6/5) \text{ A}$. Potential difference across the $2 \mu\text{F}$ capacitor is

$$2I_1 = 2 \times \frac{4}{5} = \frac{8}{5} \text{ V}$$

Energy in it is given by

$$U_1 = \frac{1}{2} \times 2 \left(\frac{8}{5}\right)^2 = (64/25) \mu\text{J}$$

Potential difference across the $3 \mu\text{F}$ capacitor is

$$3I_1 = 3 \times \frac{4}{5} = \frac{12}{5} \text{ V}$$

Energy in it is given by

$$U_2 = \frac{1}{2} \times 3 \left(\frac{12}{5}\right)^2 = (216/25) \mu\text{J}$$

Potential difference across the $3 \mu\text{F}$ capacitor is

$$3I_2 = 3 \times \frac{2}{5} = \frac{6}{5} \text{ V}$$

Energy in it is given by

$$U_3 = \frac{1}{2} \times 3 \left(\frac{6}{5}\right)^2 = (54/25) \mu\text{J}$$

Potential difference across the $7 \mu\text{F}$ capacitor is

$$7I_2 = 7 \times \frac{2}{5} = \frac{14}{5} \text{ V}$$

Energy in it is given by

$$U_4 = \frac{1}{2} \times 7 \left(\frac{14}{5}\right)^2 = (686/25) \mu\text{J}$$

Thus, whole of the energy stored in all capacitors will be dissipated after the switch is opened. So the energy dissipated is

$$U_1 + U_2 + U_3 + U_4 = 40.8 \mu\text{J}$$

For Problems 10–14

10. (2), 11. (2), 12. (3), 13. (2), 14. (1)

$$\eta = \frac{0.4 \times 4200(\theta - \theta_1)}{420 \times 4.8(\theta - \theta_1)} \times 100 = 250/3 = 83.34\%$$

$$\eta = \frac{0.4 \times 4200(\theta - \theta_1)}{420 \times 6.4(\theta - \theta_1)} \times 100 = 62.5\%$$

$$H_A = 0.2 \times 1680(\theta - \theta_1) + 0.4 \times 4200(\theta - \theta_1) = 420 \times [0.8 + 4](\theta - \theta_1)$$

$$H_B = 0.4 \times 2450(\theta - \theta_1) + 0.4 \times 4200(\theta - \theta_1) = 420 \times (2.4 + 4)(\theta - \theta_1)$$

$$\frac{H_A}{H_B} = \frac{4.8}{6.4} = \frac{3}{4}$$

$$H_A = \frac{V^2}{R_A} \times 6 \quad \dots(i)$$

$$H_B = \frac{V^2}{R_B} \times 8 \quad \dots(ii)$$

$$\frac{H_A}{H_B} = \frac{R_B}{R_A} \times \frac{3}{4} \quad \text{but} \quad \frac{H_A}{H_B} = \frac{3}{4}$$

Then $R_B = R_A$

$$14. (1) (i) \quad \frac{V^2}{R_A} \times 6 = \frac{V^2}{(R_A + R_B)^2} R_A \times t_A$$

$$\text{But } R_A = R_B \text{ or } \frac{V^2}{R_A} \times 6 = \frac{V^2}{4R_A^2} R_A \times t_A$$

$$t_A = 24 \text{ min}$$

$$(ii) \quad \frac{V^2}{R_B} \times 8 = \frac{V^2}{(R_A + R_B)^2} R_B \times t_B$$

$$\frac{8}{R_B} = \frac{R_B}{4R_B^2} t_B \text{ or } t_B = 32 \text{ min.}$$

For Problems 15–17

15. (1), 16. (2), 17. (2)

$$I \propto r^{3/2}$$

$$\frac{I_1}{I_2} = \frac{(4r)^{3/2}}{r^{3/2}} = \frac{8}{1}$$

$$P = \frac{V^2}{R} \text{ or } 2 \times 10^3 = \frac{V^2}{20}$$

$$\text{or } V = 200 \text{ V}$$

$$\eta = \frac{R}{R+r} \times 100, \text{ where } R = r, \text{ efficiency will be maximum.}$$

Matrix Match Type

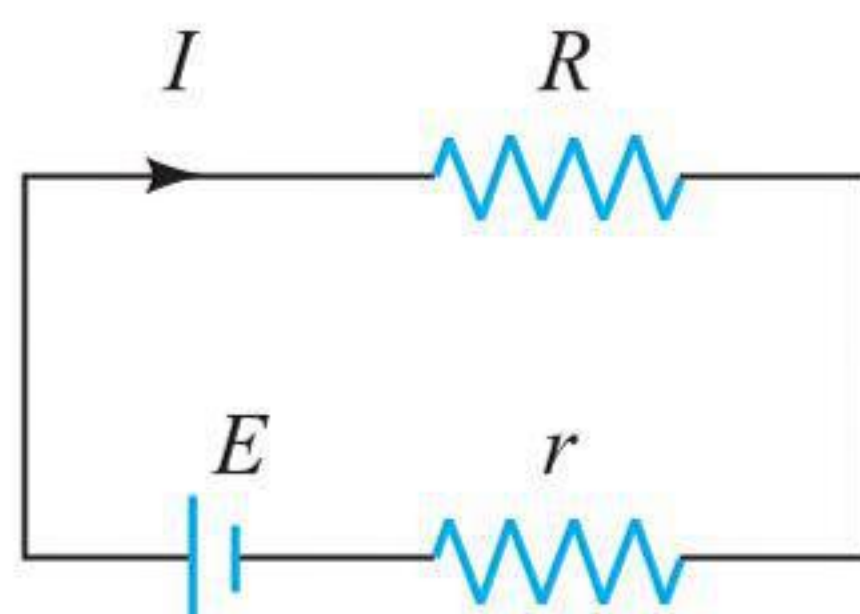
1. i. $\rightarrow$ c.; ii. $\rightarrow$ a., b., c., d.; iii. $\rightarrow$ d.; iv. $\rightarrow$ d.

$$I = \frac{E}{R+r}$$

Terminal potential difference across the cell is $V = E - Ir$. This is maximum if $I = 0$, and this is possible if $R = \infty$. Hence, i. $\rightarrow$ c.

Power transferred to R is maximum if $R = r$. In all other cases it will be somewhat less. Hence, ii. $\rightarrow$ a., b., c., d.

Power dissipated in cell is EI . This is maximum if I is maximum, for this $R = 0$. Hence, iii. $\rightarrow$ d.



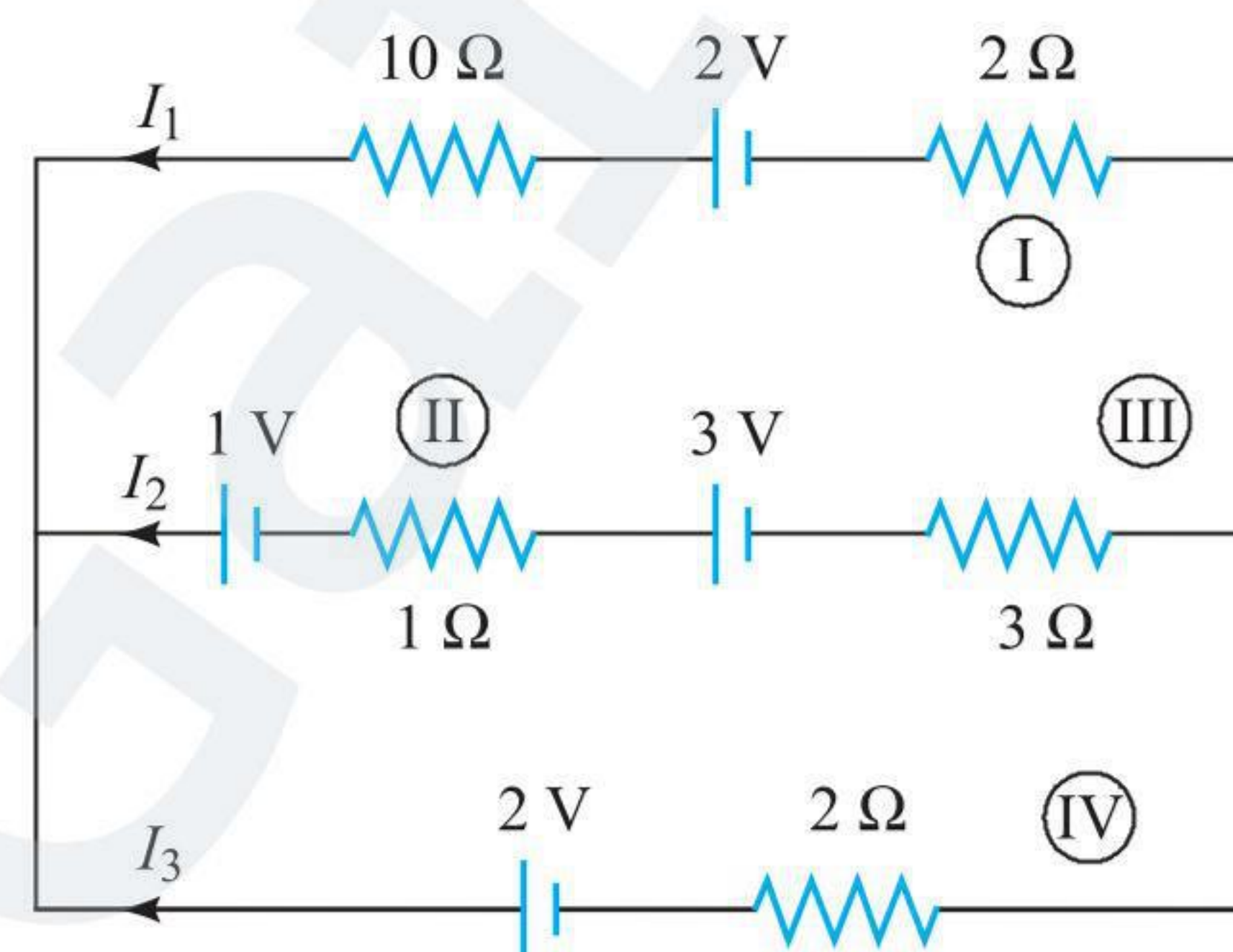
Also if I is maximum, there will be the fastest drift of ions in electrolyte in the cell. Hence, iv. $\rightarrow$ d.

2. i. $\rightarrow$ b., d.; ii. $\rightarrow$ a., c., d.; iii. $\rightarrow$ a., c., d.; iv. $\rightarrow$ b., d.

$$\text{We have } I_1 = -\frac{1}{20} \text{ A}, I_2 = \frac{7}{20} \text{ A}, I_3 = -\frac{6}{20} \text{ A.}$$

In each cell, thermal energy will be dissipated due to internal resistance whether the chemical energy of the cell is increasing or decreasing.

- (i) Cell I is getting charged, hence its chemical energy increases.
(ii) Cells II and III both are getting discharged, hence their chemical energies are decreasing. So work done by both of them is positive.
(iv) Cell IV is getting charged, hence its chemical energy increases.

3. i. $\rightarrow$ a., d.; ii. $\rightarrow$ c.; iii. $\rightarrow$ c.; iv. $\rightarrow$ c.

Charging rate means current $\left(\frac{dq}{dt} = I\right)$. So if charging rate is maximum, then current is also maximum. Hence, i. $\rightarrow$ a, d.

Charge on the capacitor is maximum in steady state. In steady state, current becomes zero. Hence, ii. $\rightarrow$ c.

Power supplied by the battery is maximum initially because current is maximum initially. At this time, charge on the capacitor is zero. Hence, iii. $\rightarrow$ c.

Difference in the power supplied by battery and power consumed in R is $EI - I^2R = EI - I(IR) = EI - IE = 0$. Hence iv. $\rightarrow$ c.

4. i. $\rightarrow$ b.; ii. $\rightarrow$ c.; iii. $\rightarrow$ c.; iv. $\rightarrow$ c.

When the switch is closed, equivalent resistance is R . After opening the switch, the equivalent resistance becomes $2R$. Hence, equivalent resistance increases. Also current through battery decreases, hence ammeter reading decreases. Current through left R also decreases. So voltmeter reading decreases and power dissipated through left R also decreases.

5. i. $\rightarrow$ b., d.; ii. $\rightarrow$ b., c.; iii. $\rightarrow$ a., d.; iv. $\rightarrow$ a., c.

$$\text{For } P_{\max} = R = 2r$$

$$P_{\max} = \left(\frac{2E}{R+2r}\right)^2 R = \frac{E^2}{2r}$$

$$\text{For } P_{\max} : R = \frac{r}{2}$$

$$P_{\max} = \left(\frac{E}{R+R/2}\right)^2 R = \frac{E^2}{2r}$$

$$P = \left(\frac{2E}{r+2r}\right)^2 r = \frac{4E^2}{9r}$$

$$P = \left(\frac{2E}{r+r/2}\right)^2 r = \frac{4E^2}{9r}$$

6. i. $\rightarrow$ d.; ii. $\rightarrow$ c.; iii. $\rightarrow$ a.; iv. $\rightarrow$ b., d.

Rated voltage is same, $P_A > P_B > P_C$

So, $R_A < R_B < R_C$

Currents through B and C are I_1 and I_2 , respectively, then current through A is $(I_1 + I_2)$. B and C have same applied voltage.

$$\frac{V_1^2}{R_B} > \frac{V_1^2}{R_C}. \text{ Power delivered to } C \text{ is minimum.}$$

Currents through A and B are I_1 and I_2 , respectively, then current through C is $(I_1 + I_2)$ and P_R is maximum for C .

$$\text{Further, } \frac{V_1^2}{R_A} > \frac{V_1^2}{R_B}. \text{ So, power delivered to } B \text{ is minimum.}$$

Current through A , B , and C is same. P_R is maximum for C and minimum for A .

Voltage is same. So, V^2/R is maximum for A and minimum for C .

Numerical Value Type

1. (4) Let resistance of heating coil be R , then

$$100 = \frac{220^2}{R} \text{ or } R = \frac{(220)^2}{100} \Omega$$

Resistance of each cut part is $R/2$. Now, power dissipated is

$$P = 2 \left(\frac{(220)^2}{R/2} \right) = 4 \frac{(220)^2}{R} = 4 \times 100 = 400 \text{ W}$$

2. (9) For three identical resistors in series, $P_s = V^2/3R$. If they are now in parallel over the same voltage,

$$P_p = \frac{V^2}{R_{eq}} = \frac{V^2}{R/3} = \frac{9V^2}{3R} = 9P_s = 9 \times (27 \text{ W})$$

3. (6) $150 = \frac{15 \times 15}{R_{eq}}$ or $R_{eq} = \frac{15 \times 15}{150} \Omega = \frac{3}{2} \Omega$

$$\text{Now, } \frac{2R}{2+R} = \frac{3}{2} \text{ or } R = 6 \Omega$$

4. (1) The resistance of four sections are shown in the figure.

Hence, the equivalent resistance R across AB is

$$\frac{1}{R} = \frac{1}{24} + \frac{1}{12} + \frac{1}{12} + \frac{1}{24} \text{ or } R = 4 \Omega$$

$$\text{Power} = V^2/R = 20^2/4 = 100 \text{ W}$$

5. (2) If the wire is connected as such across the battery, then current in wire,

$$i = \frac{V}{R} = \frac{200}{80} = 2.5 \text{ A}$$

and power obtained,

$$P = \frac{V^2}{R} = \frac{200 \times 200}{80} = 500 \text{ W}$$

The wire can carry maximum current of 5 A; therefore, to double the current, the resistance should be halved. Thus, if we divide the wire in two parts and the two parts are connected in parallel across 200 V mains supply, then the resistance of each part is 40 W; therefore, current in each wire is $200/40 = 5 \text{ A}$.

$$\text{Net resistance, } R' = \frac{R_1 R_2}{R_1 + R_2} = \frac{40 \times 40}{40 + 40} = 20 \Omega$$

and new power obtained is

$$P_{\max} = \frac{V^2}{R} = \frac{200 \times 200}{20} = 2000 \text{ W}$$

Thus, the maximum power is 2000 W and this is obtained when wire is cut in two halves and they are connected in parallel across the given supply.

6. (6) $i = 2 \text{ A}$, $R = 25 \text{ W}$, $t = 1 \text{ min} = 60 \text{ s}$

$$\text{Heat developed} = i^2 R t = (2 \times 2 \times 25 \times 60) = 100 \times 60 \text{ J} = 6000 \text{ J}$$

7. (2) As discussed in maximum power transfer theorem the maximum power which can be transferred is given as

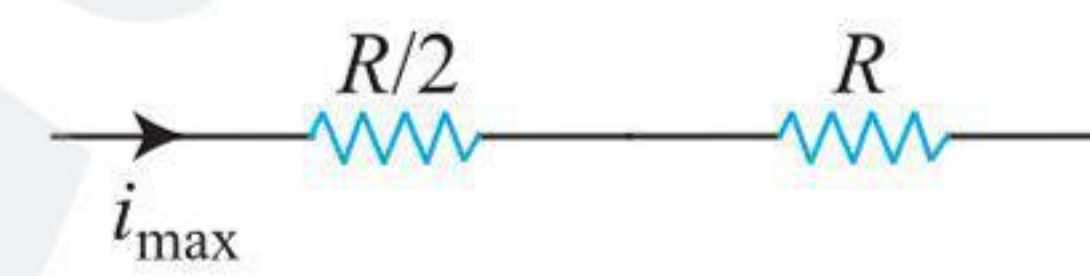
$$P_{\max} = \frac{E^2}{4r}$$

Here E and r are the equivalent EMF and internal resistance of the equivalent battery when the given batteries are connected in series as in that case EMF will be maximum which gives

$$P_{\max} = \frac{(4)^2}{4 \times 2} = 2 \text{ W}$$

8. (54) $P = i^2 R$

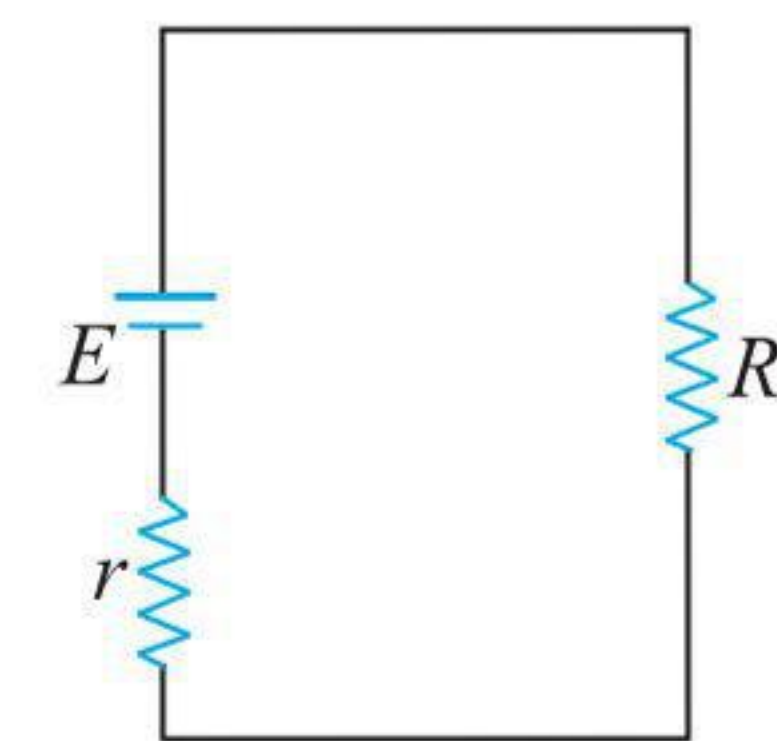
$$\therefore i_{\max} = \sqrt{\frac{P_{\max}}{R}} = \sqrt{\frac{36}{2.4}} = \sqrt{15} \text{ A}$$



$$\text{Total maximum power} = (i_{\max})^2 \left(\frac{3R}{2} \right)$$

$$= (15)(1.5)(2.4) = 54 \text{ W}$$

9. (0.3) Figure shows the situation described in question. Here r is the internal resistance of the battery and E its EMF.



Power supplied by the battery in this case is given as

$$P = \frac{E^2}{R + r}$$

Substituting the values gives

$$0.4 = \frac{(6)^2}{60 + r} \Rightarrow r = 30 \Omega$$

Maximum power is dissipated in the circuit when external resistance is equal to net internal resistance which gives

$$R = r \Rightarrow R = 30 \Omega$$

Total power supplied by the battery under this condition is given as

$$P_{\text{Total}} = \frac{E^2}{R + r} = \frac{(6)^2}{30 + 30} \Rightarrow P_{\text{Total}} = 0.6 \text{ W}$$

Out of this 0.6 W half of the power is dissipated in R and half is dissipated in r . Therefore, maximum power dissipated in R is 0.3 W.

10. (384) The resistance of heater coil is given as

$$R = \frac{V^2}{P} = \frac{(200)^2}{500} = \frac{40000}{500} = 80 \Omega$$

When connected across 160 V supply, the power consumed by heater will be given as

$$P' = \frac{V^2}{R} = \frac{(160)^2}{80} = \frac{160 \times 160}{80}$$

$$\Rightarrow P' = 320 \text{ W}$$

Total heat produced in 20 minute (1200 s) is given as

$$H = P't$$

$$\Rightarrow H = 320 \times 1200 \Rightarrow H = 384000 = 384 \text{ kJ}$$

11. (0.3) When a current is drawn from the battery, its potential difference across the terminals decreases which is given as

$$2.0 = 2.6 - Ir$$

$$\Rightarrow r = 0.6 \Omega$$

If external resistance is R , we use

$$R + r = \frac{2.6}{1} = 2.6 \Omega \Rightarrow R = 2.6 - 0.6 = 2.0 \Omega$$

Total thermal power generated in the battery is given as

$$P_{\text{Thermal}} = I^2 r = 1^2 \times 0.6 = 0.6 \text{ W}$$

Total electrical power supplied by the battery is given as

$$P_{\text{total}} = 1 \times 2 = 2 \text{ W}$$

$$\text{Hence required ratio } \frac{P_{\text{Thermal}}}{P_{\text{total}}} = \frac{0.6}{2} = 0.30$$

$$12. (20) P_{\text{initial}} (= P_2) = \frac{\varepsilon^2 R'}{(R + R')^2}$$

$$P_{\text{final}} (= P_2) = \frac{2\varepsilon^2 (R'/2)}{(R + R'/2)^2}$$

$$\text{Since, } P_1 = P_2$$

$$\text{Or } (R + R')^2 = 2 \left(R + \frac{R'}{2} \right)^2$$

$$\text{Or } (R + R') = \left(R + \frac{R'}{2} \right) \sqrt{2}$$

$$\text{Or } R = \frac{R'}{\sqrt{2}} = \frac{20\sqrt{2}}{\sqrt{2}} = 20 \Omega$$

13. (44.4) Initial potential energy stored in the capacitor,

$$U = \frac{1}{2} CV^2$$

$$\Rightarrow U = \frac{1}{2} \times 5 \times 10^{-6} \times (200)^2 = 0.1 \text{ J.}$$

When the switch is shifted from contact 1 to 2 capacitor starts discharging and during discharging this 0.1 J, will distribute in direct ratio of resistance thus heat dissipated in 400Ω resistance is given as

$$H_{400\Omega} = \frac{400}{400 + 500} \times 0.1$$

$$\Rightarrow H_{400\Omega} = 44.4 \times 10^{-3} \text{ J.}$$

$$\Rightarrow H_{400\Omega} = 44.4 \text{ mJ}$$

$$14. (25) \text{ The resistance } R = \frac{V^2}{P} = \frac{(250)^2}{100} = 625 \Omega$$

This is a balanced Wheatstone bridge $\Rightarrow i' = 0$

$$\text{Then, } i = \frac{\varepsilon}{R + r} = \frac{260}{625 + 25} = \frac{2}{5} \text{ A}$$

$$P = \left(\frac{i}{2} \right)^2 R = \left(\frac{1}{2} \right)^2 (625) = 25 \text{ W}$$

$$15. (7.5) \text{ Current in first case, } i = \frac{E}{5 + x} = \frac{20}{5 + x}$$

Power supplied to the resistance,

$$P = i^2 R = \frac{400}{(5 + x)^2} \times x \quad \dots(i)$$

$$\text{In second case, current in the circuit, } i' = \frac{E}{5 + \left(\frac{6x}{x + 6} \right)}$$

Power supplied by the battery in this case,

$$P' = i'^2 R' = \frac{400}{\left[5 + \left(\frac{6x}{x + 6} \right) \right]^2} \frac{6x}{(6 + x)} \quad \dots(ii)$$

As the power supplied in two cases are equal, from (i) and (ii)

$$\frac{400x}{(5 + x)^2} = \frac{400}{\left[5 + \left(\frac{6x}{x + 6} \right) \right]^2} \times \frac{6x}{(6 + x)} \Rightarrow x = 7.5 \Omega$$

Archives

JEE Advanced

Single Correct Answer Type

1. (4) Power $\propto 1/R$

Multiple Correct Answers Type

1. (1),(4)

$$24 - 2 \times 10^3 I - 6 \times 10^3 (I - i) = 0$$

$$24 - 2 \times 10^3 I - 1.5 \times 10^3 i = 0$$

$$\text{Hence, } I = 7.5 \text{ mA}$$

$$i = 6 \text{ mA}$$

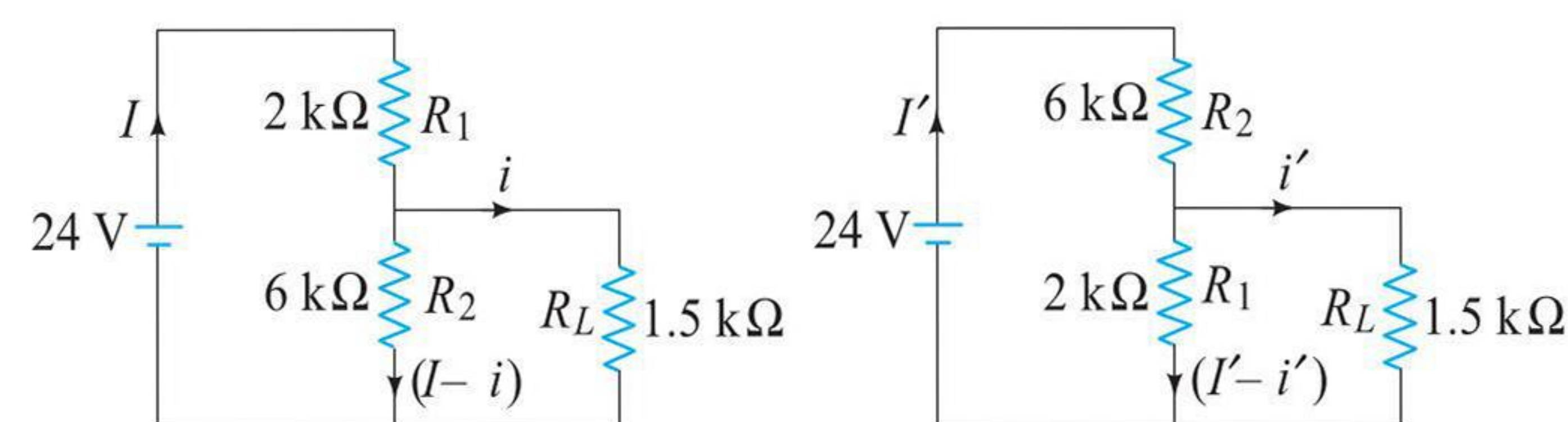
$$24 - 6 \times 10^3 I' - 2 \times 10^3 (I' - i') = 0$$

$$24 - 6 \times 10^3 I' - 1.5 \times 10^3 i' = 0$$

$$I' = 3.5 \text{ mA}$$

$$i' = 2 \text{ mA}$$

$$\frac{P_1}{P_2} = \frac{6^2}{2^2} = 9$$



2. (2),(4)

Heat required to increase the temperature of water by 40 K.

$$\Delta Q = mc\Delta T = \left(\frac{\varepsilon^2}{R} \right) t$$

$$\Rightarrow \Delta Q = 0.5 \times c \times 40 = \left(\frac{\varepsilon^2}{R} \right) \times 4 \text{ min}$$

$$\text{Where } R = \frac{\rho L}{\left(\frac{\pi d^2}{4} \right)}$$

The resistance of wire of length L and diameter $2d$.

$$R_1 = \frac{\rho L}{\pi(2d)^2/4} = \frac{R}{4}$$

$$\text{If resistances are in series, } R_{\text{eq}} = \frac{R}{4} + \frac{R}{4} = \frac{R}{2}$$

$$\text{Heat required, } \Delta Q = \left(\frac{\varepsilon^2}{R} \right) 4 = \frac{\varepsilon^2}{\left(\frac{R}{2} \right)} \times t$$

Which gives $t = 2 \text{ min}$

$$\text{If resistances are in parallel, } R_{\text{eq}} = \frac{\frac{R}{4} \times \frac{R}{4}}{\frac{R}{4} + \frac{R}{4}} = \frac{R}{8}$$

$$\text{Heat required, } \Delta Q = \left(\frac{\varepsilon^2}{R} \right) 4 = \frac{\varepsilon^2}{\left(\frac{R}{8} \right)} \times t$$

Which gives $t = 0.5 \text{ min}$

Numerical Value Type

$$1. (4) J_1 = \left(\frac{2E}{R+2} \right)^2 R$$

$$J_2 = \left(\frac{E}{R+1/2} \right)^2 R \text{ since } J_1 / J_2 = 2.25$$

$$R = 4 \Omega$$

